

Linear Programming of Graph Squares

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Abstract

The square of a graph is obtained by joining every pair of vertices whose distance is at most two. Given a graph G , the graph square root problem asks whether there exists a graph H such that $G = H^2$. Whilst this problem admits polynomial-time algorithms for restricted classes of root graphs, the general decision problem is *NP*-complete.

In this thesis, rather than studying the construction problem directly and using reduction techniques to ascertain polynomial-time algorithms, we formulate the problem as an integer program based on the relationship between the adjacency matrices of a root H and its square G . By constructing the corresponding linear relaxation, the problem can then be situated within polynomial time through the use of interior-point methods.

The seminal result for the problem is the work by Ross and Harary, who approached the problem by restricting the root graphs to trees. By analyzing the number of cliques which each vertex in the graph square is situated in, they gave a polynomial-time algorithm for finding the unique tree root up to isomorphism. In contrast, our formulation provides an optimization-based approach that can be iteratively strengthened through the addition of further constraints. These constraints refine the linear relaxation, enabling tree roots to be represented by integral solutions to the linear program, while graphs with no possible roots are identified through fractional or infeasible solutions. Rather than relying on heuristic such as rounding, our approach exploits structural properties of the graph square and its root to obtain an exact solution.

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Chapter 1

Introduction

1.1 Background

In 1949, Leon Festinger proposed a novel method in the field of social psychology for analyzing relationships among a number of people present [FBB23]. Classically, this was represented using a sociogram, a visual tool that maps out the relationships between people. However, Forsyth and Katz introduced the use of a matrix to represent the sociometric pattern, with rows and columns corresponding to individuals and containing 1s for choices and blank diagonals. Festinger noticed that his structure, reminiscent of the adjacency matrix defined by König in [Kön36], also had the remarkable property that the action of squaring a matrix

$$A_{rc}^2 = A_{1c}A_{r1} + \cdots + A_{nc}A_{rn}^1$$

represented the number of “two-step” connections, which can then be used to denote “three-step” connections between any two people

$$A_{rc}^3 = A_{1c}A_{r1}^2 + \cdots + A_{nc}A_{rn}^2,^2$$

such that higher steps can be easily generated. In this manner, higher powers of the adjacency matrix encode the number of walks of increasing length between pairs of individuals, allowing densely connected groups of people, known as cliques, to be identified.

In the process of improving Festinger’s clique detection methods, Ross and Harary showed that complete graphs may be realized as squares of stars by considering powers of such matrices [RH60]. They introduced a structural characterization of the square of a graph in terms of clique intersections, “multicliquial” vertices, and so called “neighborly” properties, proving that a graph is the square of a tree if and only if

- i) every vertex is every vertex is adjacent to at least one other vertex,
- ii) no two maximal cliques intersect in more than two vertices, and

¹ A_{rc}^2 is more naturally written as $\sum_{k=1}^n A_{ck}A_{kr}$ using the commutativity of multiplication $A_{1c}A_{r1} = A_{c1}A_{1r}$, which intuitively gives the notion of paths from c to r via intermediary k

² A_{rc}^3 is more naturally written as $\sum_{k=1}^n A_{ck}A_{kr}^2$, which intuitively counts the number of paths from c to r that go through an intermediate vertex k , followed by any two-step path from k to r .

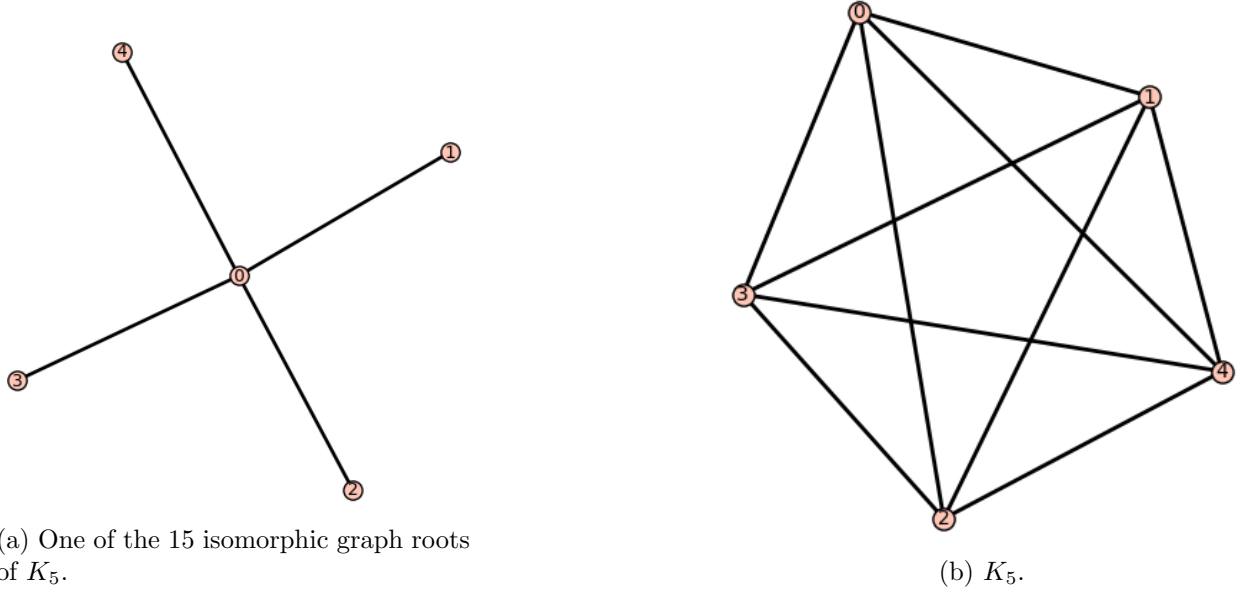


Figure 1.1: The star graph $K_{1,4}$ and its square K_5 , which is a complete graph since every vertex in the root .

- iii) the number of maximal clique pairs intersecting in two vertices is exactly one less than the total number of maximal cliques.

Using these properties, they constructed an explicit algorithm that recovers the underlying tree for a given graph square. Their work has become the foundational reference for the study of graph squares and is universally cited in subsequent literature on the subject, and will be explored in more detail later.

Since then, the problem of finding the square root of a graph has extended beyond sociology and is a well-studied field in both mathematics and computer-science, with direct applications in distributed computing (e.g., see [PA23]), network design and computational biology (e.g., see [Zho+25]). This is in part because many structural properties of graph roots are first established in the square case before being extended to the more general k -th power setting (e.g., see [HKT67]).

Research in this area has primarily focused around an undirected graph H and its square $G = H^2$ which is obtained by adding an edge between every pair of vertices whose distance in H is at most two. This two-step undirected graph square has been studied extensively with much of the focus being on the complexity of the decision problem - given a specific graph square G does there exist a root graph H such that $H^2 = G$. Motwani and Sudan showed that deciding if a graph admits a root is NP -complete. Subsequent studies in the area have restricted the decision problem in two ways, either by considering various properties on the root graph or by considering properties on the graph square.

1.2 Literature

Focusing first on special undirected graph classes, the aforementioned work by Ross and Harary in 1960 gives an algorithm for constructing a tree root by using characterizations of the graph square. As part of this work they proved tree square roots are unique up to isomorphism. This was the first major contribution to the study of graph roots and provided fundamental structural

characteristics of graph squares, laying the groundwork for later research on graph root problems and their applications.

Fleischner in 1974 showed that as long as a graph is 2-connected its graph square will have a Hamiltonian cycle. This established a broad condition guaranteeing Hamiltonicity in graph squares which is used in many subsequent works such as [BMR14].

Lau and Corneil in 2004 provided an algorithm for constructing square roots from an interval square in polynomial time, and showed that the decision problem for finding roots of chordal graphs, roots of split graphs, and chordal roots of graphs are *NP*-complete [LC04]. Furthermore, Lau gave a linear-time algorithm for constructing square roots of bipartite graphs, and proved that deciding whether a bipartite graph has a cube root is *NP*-complete [Lau06]. Le and Tuy in 2007 generalized tree-root results further by giving a linear time recognition of roots of block graphs [LT07].

Generalizing the notion of graph squares, in 1967 Mukhopadhyay provided necessary and sufficient conditions for a connected graph to have a square root [Muk67], whilst Geller in 1970 likewise found necessary and sufficient conditions for a digraph to have a square root [Gel70].

In 1994 Motwani and Sudan established the complexity of the square problem for the generalized undirected case and showed that the problem of determining if a given graph has a square root is a *NP*-complete problem via a polynomial reduction to 3-CNF Not-All-Equal SAT [MS94].

Their work led to joint work by Farzad, Le, Tuy, Lau and Karimi in 2009 and 2012, in which they established a partial dichotomy theorem for the complexity of determining if a given undirected graph of girth g has a square root [Far+09] which was informally completed under [FK12]. In 2013, Cochefert, Couturier, Golovach, Kratsch, and Paulusma gave an algorithm for constructing a square root with minimal number of edges from undirected graphs with max degree ≤ 6 in polynomial time and an algorithm for constructing maximal edge square roots in exponential time [Coc+13].

In recent years, Duffy provided a full classification for the undirected square of oriented graphs satisfying max degree ≤ 3 and girth ≥ 4 [Duf25].

1.2.1 Overview

In addition to providing a general overview of the research direction in this section we first standardise the definitions that will be used for the rest of this thesis. We then introduce fundamental theory on the three areas of: complexity theory, graph theory and mathematical optimization that are used in the various algorithmic approaches to the decision problem.

The first construction we examine is the reduction used by Motwani and Sudan in showing that the decision problem determining whether a given graph G admits a square root is *NP*-complete [MS94]. Next, we introduce the polynomial time algorithm given by Ross and Harary for constructing a tree root from a given square in detail. Finally, we formulate the problem as an integer program.

In Chapter 3 we present the ideas of linear relaxation and discuss the results of each program and their validity. To obtain these results, we conducted an exhaustive enumeration on all non-isomorphic connected graphs up to $n = 8$ vertices. For each such graph, we formulated the corresponding linear program and solved it using a standard LP solver using interior point method. The outcomes of these computations are categorized as integral, fractional, or infeasible and are analyzed.

Finally, we introduce several augmentations to the linear program, consisting of constraints that encode structural bounds on both the graph square and its underlying root. In particular, we show that by adding specific constraints to the LP, we can ensure that the square of all trees will return a feasible integer solution. This augmentation ensures that the LP returns a tree root in polynomial time, whilst ensuring that all squares with roots remain LP feasible.

1.3 Definitions and Notations

1.3.1 Complexity Theory

Complexity theory studies the classification of computational problems and explores the amount of resources required to solve these problems. Complexity Theory considers both time and space as critical resources and is standardized across various different problems.

Definition 1.3.1 (Big- O Notation). Let $f, g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$. The set $O(g(n))$ is defined as

$$O(g(n)) = \{f \mid \exists c > 0, n_0 \in \mathbb{N} \text{ such that } f(n) \leq cg(n) \text{ for all } n \geq n_0\}.$$

We write $f(n) = O(g(n))$ as shorthand for $f(n) \in O(g(n))$.

Using this notation, we can classify algorithms based on their runtime.

Definition 1.3.2 (Polynomial-Time Algorithm). An algorithm is said to run in **polynomial time** if there exists a constant $c \geq 0$ such that its run time on inputs of size n is $O(n^c)$.

Polynomial-time algorithms are important because they are generally regarded as efficient and computationally “simple”. Consequently, the class of problems that admit polynomial-time algorithms forms the basis for the complexity class **P**.

Definition 1.3.3 (Decision Problem). A **decision problem** is a computational problem whose output is either **Yes** or **No** for every valid input instance.

Definition 1.3.4 (Class **P**). The complexity class **P** consists of all decision problems that can be solved by a deterministic algorithm in polynomial time.

Unfortunately many important decision problems are not known to admit such polynomial-time algorithms and hence need additional classification on their “hardness”. For example, determining whether a graph is 3-colourable is a well-known problem for which no polynomial-time algorithm is known, despite the fact that a proposed 3-colouring can be checked efficiently. This motivates the study of complexity classes based not only on how efficiently a solution can be found, but also on how a proposed solution can be verified efficiently.

Definition 1.3.5 (Certificate). Let L be a decision problem. A **certificate** for an instance x of L is a finite string c that provides evidence that x is a **Yes**-instance of L . A certificate is said to be **polynomially verifiable** if there exists a deterministic algorithm that for given input (x, c) , can determine whether c is a valid certificate for x in time polynomial in $|x|$.

Definition 1.3.6 (Class **NP**). The complexity class **NP** consists of all decision problems for which every **Yes**-instance admits a certificate that is polynomially verifiable.

Since every decision problem in **P** can be solved in polynomial time, a solution can also be verified in polynomial time,

$$\mathbf{P} \subseteq \mathbf{NP}.$$

Whether this inclusion is strict remains one of the most important open questions in theoretical computer science - the **P** vs **NP** problem. As **NP** is a large class of decision problems, the relative difficulty of problems in **NP** is compared by using reduction.

Definition 1.3.7 (Polynomial-Time Reduction). *Let A and B be decision problems. A **polynomial-time reduction** from A to B , denoted $A \leq_p B$, is a polynomial-time computable function f such that, for every instance x of A ,*

$$x \in A \iff f(x) \in B.$$

This notion of reduction allows the computational difficulty of problems to be compared. In particular, a problem is **NP-hard** if every problem in **NP** can be reduced to it via a polynomial time reduction.

Definition 1.3.8 (Class NP-Hard). *A decision problem B is **NP-hard** if every problem $A \in \mathbf{NP}$ satisfies $A \leq_p B$. The collection of all **NP-hard** problems is denoted by the class **NP-hard**.*

Importantly, an **NP-hard** problem may not necessarily lie in **NP**. If it does, it is called **NP-complete**.

Definition 1.3.9 (Class NP-Complete). *A decision problem B is **NP-complete** if $B \in \mathbf{NP}$, and B is **NP-hard**. The collection of all **NP-complete** problems is denoted by the class **NP-complete**.*

One can establish that a problem is **NP-complete** by showing that it belongs to **NP** and that a known **NP-complete** problem can be reduced to it in polynomial time.

The first known **NP-complete** problem was established by the Cook-Levin theorem and is the Boolean satisfiability problem, or SAT.

Definition 1.3.10 (Boolean Satisfiability Problem). *The **Boolean satisfiability problem (SAT)** asks whether there exists an assignment of truth values to the variables of a given Boolean formula such that the formula evaluates to true. A Boolean formula is said to be **satisfiable** if such an assignment exists.*

Any Boolean formula can be transformed into an equisatisfiable formula typically in linear time using the **Tseytin transformation** by introducing additional variables. This transformation preserves satisfiability while ensuring only linear growth in the size of the resulting conjunctive normal form Boolean formula.

Definition 1.3.11 (Conjunctive Normal Form). *A Boolean formula is in **conjunctive normal form (CNF)** if it is expressed as a conjunction of clauses, where each clause is a disjunction of literals. That is, a CNF formula has the form*

$$C_1 \wedge C_2 \wedge \cdots \wedge C_m, \quad \text{where each clause } C_i \text{ is of the form } C_i = \ell_{i1} \vee \ell_{i2} \vee \cdots \vee \ell_{ik_i}$$

and each ℓ_{ij} is either a Boolean variable or its complement, denoted with $\bar{\ell}_{ij}$.

A further restriction is obtained by requiring each clause in a CNF formula to contain exactly three literals, which can be done using 3-CNF transformation, to give the following problem.

Definition 1.3.12 (3-SAT). *The **3-SAT** problem asks whether there exists an assignment of truth values to the variables of a Boolean formula in conjunctive normal form such that every clause contains exactly three literals and the formula evaluates to true. It is formally structured as*

Instance: A Boolean formula φ in conjunctive normal form in which each clause contains exactly three literals.

Question: Does there exist an assignment of truth values to the variables of φ such that every clause is satisfied?

The decision problem **3-SAT** is a well-known **NP-complete** problem. It is commonly used as a starting point for proving the **NP-completeness** of other decision problems. In particular, we will reduce 3-SAT to the **SQUARE** problem to establish that the latter is **NP-complete**.

1.3.2 Graph Theory

Informally, a graph is relational structure that captures relationships between discrete objects.

Definition 1.3.13 (Graph). A **graph** G is an ordered pair (V, E) , where V is a set of vertices and E is a set of unordered pairs of distinct vertices in V . Each element of E is called an **edge** of G .

The number of vertices and edges in a graph provide basic measures of its order and size.

Definition 1.3.14 (Order and Size). The **order** of a graph G , denoted by $|G|$, is the number of vertices in G , while the **size** of G is the number of edges. Thus, if $V_G = \{v_1, \dots, v_n\}$ and $E_G = \{e_1, \dots, e_m\}$, then $|G| = n$ and $|E_G| = m$.

For a vertex v , the number of edges incident to it is its degree.

Definition 1.3.15 (Degree). The **degree** of a vertex v in a graph G , denoted by $\deg_G(v)$, is the number of edges incident to v . The minimum and maximum degrees of G are denoted by

$$\delta(G) = \min_{v \in V_G} \deg_G(v) \quad \text{and} \quad \Delta(G) = \max_{v \in V_G} \deg_G(v),$$

respectively.

The degree of a vertex can also be described through the vertices to which it is adjacent to.

Definition 1.3.16 (Neighbourhood). For a vertex $v \in V_G$, the **open neighbourhood** of v denoted by $N_G(v)$, is the set of vertices adjacent to v in G . The **closed neighbourhood** of v includes vertex v and is denoted by

$$N_G[v] = N_G(v) \cup \{v\}.$$

Connections between vertices in a graph can be described using walks, paths and cycles.

Definition 1.3.17 (Walk). A **walk** from a vertex v_i to a vertex v_j in a graph G is an alternating sequence of vertices and edges

$$W = \langle v_i, e_1, v_1, e_2, \dots, e_k, v_j \rangle,$$

where each edge e_r has endpoints given by the vertices immediately preceding and following it in the sequence. The number of edges in W is called the **length** of the walk.

Definition 1.3.18 (Path). A **path** is a walk in which no vertex is repeated. A path containing p vertices is denoted by P_p .

Definition 1.3.19 (Cycle). A **cycle** is a closed path in which the first and last vertices are the same and all other vertices are distinct. A cycle containing c vertices is denoted by C_c .

The length of a shortest cycle is an important structural parameter of a graph.

Definition 1.3.20 (Girth). The **girth** of a graph G , denoted by $g(G)$, is the length of the shortest cycle in G . If G contains no cycle, its girth is ∞ .

A connected graph with infinite girth is called a tree.

Definition 1.3.21 (Tree). A **tree** is a connected graph that contains no cycles.

The length of a shortest path provides a natural measure of the distance between two vertices.

Definition 1.3.22 (Distance). The **distance** between two vertices u and v on the graph G , denoted by $d_G(u, v)$, is the number of edges of the shortest path between vertices u and v in G . If no such path exists, $d_G(u, v) = \infty$.

Definition 1.3.23 (Diameter). The **diameter** of a graph G is

$$\text{diam}(G) = \max_{u, v \in V_G} d_G(u, v).$$

For a disconnected graph, the diameter is ∞ .

If a graph has finite diameter it is connected.

Definition 1.3.24 (Connected Graph). A graph G is **connected** if there exists a path between every pair of vertices in G . Otherwise, G is **disconnected**.

A graph can be further categorized according to whether multiple edges between a vertex and itself are permitted, whether there are infinite number of vertices and edges, whether the edges represent an unordered or ordered relationship between their endpoints, and whether the vertices are labelled or unlabelled.

Definition 1.3.25 (Simple Graph). A **simple graph** $G = (V, E)$ is a graph in which no two distinct edges connect the same pair of vertices and no edge connects a vertex to itself.

Definition 1.3.26 (Finite Graph). A **finite graph** is a graph $G = (V, E)$ for which both V and E are finite sets. A graph that is not finite is called an **infinite graph**.

Definition 1.3.27 (Directed Graph). A **directed graph** $G = (V, E)$ is a graph in which each edge $e \in E$ is an ordered pair of distinct vertices in V . Thus, (u, v) represents an edge directed from u to v , where (u, v) may not necessarily be equal to (v, u) .

An **undirected graph** $G = (V, E)$ is a graph in which each edge $e \in E$ is an unordered pair of distinct vertices in V . Thus, $\{u, v\} = \{v, u\}$ for every edge between vertices u and v .

Definition 1.3.28 (Labelled Graph). A **labelled graph** is a graph $G = (V, E)$ together with a labelling function $\ell : V \rightarrow L$, where L is a set of labels. A graph is **unlabelled** when no such vertex labels are specified.

In this thesis, we exclusively study finite, simple, undirected, unlabelled graphs. However, the constructions and algorithms may use additional structure, including labels or multiple edges, which must be analysed and restricted to their corresponding simple, unlabelled counterparts.

In particular, two graphs may have different vertex sets while nevertheless having the same underlying structure. This motivates the following notion of isomorphism.

Definition 1.3.29 (Graph Isomorphism). Let G and H be graphs. An **isomorphism** from G to H is a bijection

$$f : V_G \rightarrow V_H \quad \text{such that} \quad uv \in E_G \text{ if and only if } f(u)f(v) \in E_H.$$

We say that G is **isomorphic** to H , written $G \cong H$, if there exists an isomorphism from G to H .

The isomorphism relation is an equivalence relation on the set of graphs, and an **isomorphism class** is an equivalence class of graphs under this relation. Thus, graphs within the same isomorphism class have the same structure up to a relabelling of their vertices.

We will also consider graphs obtained by restricting the vertices and edges of another graph.

Definition 1.3.30 (Subgraph). A graph $H = (V_H, E_H)$ is a **subgraph** of a graph $G = (V_G, E_G)$ if

$$V_H \subseteq V_G \quad \text{and} \quad E_H \subseteq E_G.$$

For $S \subseteq V_G$, the **induced subgraph** of G on S , denoted by $G[S]$, is the graph with vertex set S and edge set

$$E(G[S]) = \{uv \in E_G : u, v \in S\}.$$

One of the most important types of graphs is the complete graph, and its subgraph counterpart.

Definition 1.3.31 (Complete Graph). A graph on n vertices is **complete** if every pair of distinct vertices is adjacent. The complete graph on n vertices is denoted by K_n .

Definition 1.3.32 (Clique). A **clique** of a graph G is a subset $S \subseteq V_G$ such that the induced subgraph $G[S]$ is complete. The **clique number** of G , denoted by $\omega(G)$, is the size of a largest clique in G . A clique containing k vertices is denoted by K_k .

Definition 1.3.33. A **multicliqual vertex** is a vertex that belongs to multiple maximal cliques, an important distinction from just cliques since any vertex belonging to a clique of size n naturally belongs to a clique of size $n - 1$. A **unicliqual vertex** is a vertex that belongs to only one maximal clique.

In addition to being specified using a set of vertices and a set of edges, a graph can also be represented using a matrix.

Definition 1.3.34 (Adjacency Matrix). The **adjacency matrix** of graph H is denoted by $A_H \in \{0, 1\}^{n \times n}$ where

$$(A_H)_{ij} = \begin{cases} 1, & \text{if } ij \in E_H, \\ 0, & \text{otherwise,} \end{cases}$$

The degrees of the vertices can similarly be represented by a diagonal matrix.

Definition 1.3.35 (Degree Matrix). The **degree matrix** of graph H is denoted by $D_H \in \mathbb{R}^{n \times n}$ where

$$(D_H)_{ij} = \begin{cases} \deg_H(i), & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases}$$

In our discussion of graphs represented as matrices, we will find the following notation useful.

Definition 1.3.36 (Indicator Function). Let $\mathbb{1}_{\geq 1} : \mathbb{R}^{n \times n} \rightarrow \{0, 1\}^{n \times n}$ denote the entrywise **indicator function**, defined by

$$\mathbb{1}_{\geq 1}(M)_{ij} = \begin{cases} 1, & \text{if } M_{ij} \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

We now introduce the graph-theoretic objects central to this thesis.

Definition 1.3.37 (Graph Root). Let the **graph root** $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set.

Definition 1.3.38 (Square of a Graph). The **square of a graph** H , denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \quad (1.1)$$

That is, two distinct vertices are adjacent in H^2 if and only if they are connected by a path of length one or two in H . This is the standard definition of the square of a graph.

Definition 1.3.39 (Graph Square). The **graph square** of H is the undirected graph defined on the same vertex set $G = (V, E_G)$. Define the adjacency matrix of G using the adjacency and degree matrices of H

$$A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \quad (1.2)$$

1.3.3 Mathematical Optimization

Mathematical optimization studies the problem of finding the best solution from all feasible solutions which are defined by a collection of constraints. An optimization problem usually consists of decision variables, an objective function to be maximized or minimized, and a set of constraints restricting the feasible values of the variables.

Definition 1.3.40 (Integer Program). An **integer program** (LP) is an optimization problem of the form

$$\begin{aligned} \max/\min \quad & c^\top x \\ \text{subject to} \quad & Ax \leq b, \\ & x \in \mathbb{Z}^n, \end{aligned}$$

where $x \in \mathbb{Z}^n$ is the vector of decision variables, $c \in \mathbb{R}^n$ is the objective coefficient vector, $A \in \mathbb{R}^{m \times n}$ is the constraint matrix, and $b \in \mathbb{R}^m$ is the constraint vector.

A linear relaxation is obtained by removing the integrality requirements from an integer program while retaining its linear objective function and constraints.

Definition 1.3.41 (Linear Relaxation). Let IP be an integer program. The **linear relaxation** of IP is the linear program obtained by removing the requirement that its variables take integer values.

Definition 1.3.42 (Linear Program). A **linear program** (LP) is an optimization problem of the form

$$\begin{aligned} \max/\min \quad & c^\top x \\ \text{subject to} \quad & Ax \leq b, \\ & x \in \mathbb{R}^n, \end{aligned}$$

where $x \in \mathbb{R}^n$ is the vector of decision variables, $c \in \mathbb{R}^n$ is the objective coefficient vector, $A \in \mathbb{R}^{m \times n}$ is the constraint matrix, and $b \in \mathbb{R}^m$ is the constraint vector.

Definition 1.3.43 (Feasible Solution). A **feasible solution** to an optimization problem is an assignment of values to its decision variables that satisfies all of its constraints. The set of all feasible solutions is called the **feasible set** and is denoted by $\mathcal{F}$. Geometrically, $\mathcal{F}$ is the subset of the decision-variable space satisfying all of the constraints and is referred to as the **feasible region**.

Among the feasible solutions, an optimal solution is one that achieves the best value of the objective function.

Definition 1.3.44 (Optimal Solution). An **optimal solution** is a feasible solution that maximizes or minimizes the objective function, depending on the optimization problem. The corresponding value of the objective function is called the **optimal value** and is denoted by OPT .

When an integer program is relaxed to a linear program, every feasible integer solution remains feasible for the relaxation. Consequently, if $\mathcal{F}_{IP}$ and $\mathcal{F}_{LP}$ denote the feasible sets of the integer program and its linear relaxation respectively, then

$$\mathcal{F}_{IP} \subseteq \mathcal{F}_{LP}.$$

Geometrically, $\mathcal{F}_{IP}$ is represented by the feasible integer points in the decision-variable space, whereas $\mathcal{F}_{LP}$ is the polyhedron defined by the relaxed constraints and may contain additional points.

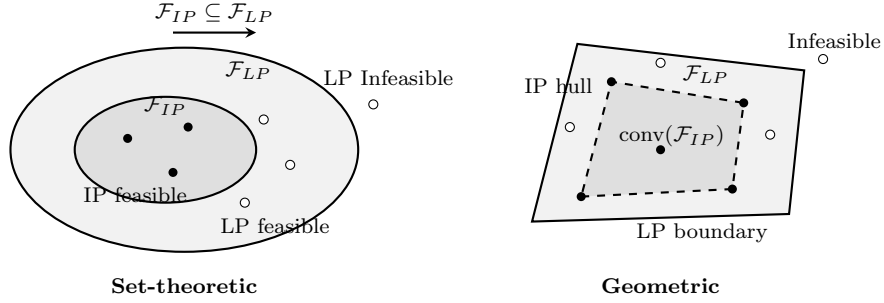


Figure 1.2: Set-theoretic and geometric relationships between an integer program and its linear relaxation.

For a maximization problem, this set inclusion implies $\text{OPT}_{IP} \leq \text{OPT}_{LP}$, so the linear relaxation provides an upper bound on the optimal value of the integer program. For a minimization problem, $\text{OPT}_{LP} \leq \text{OPT}_{IP}$, so the linear relaxation provides a lower bound.

Definition 1.3.45 (Integer Hull). *Let $\mathcal{F}_{IP}$ denote the feasible set of an integer program. The **integer hull** is the convex hull of the feasible integer solutions, denoted by $\text{conv}(\mathcal{F}_{IP})$. That is, the integer hull is the set of all convex combinations of points in $\mathcal{F}_{IP}$.*

Since the feasible region of a linear program is convex, the integer hull is contained in the feasible region of its linear relaxation. Hence,

$$\mathcal{F}_{IP} \subseteq \text{conv}(\mathcal{F}_{IP}) \subseteq \mathcal{F}_{LP}.$$

The integer hull is the smallest convex set containing all feasible integer solutions. Thus, geometrically, the gap between the integer hull and the LP feasible region represents the additional solutions admitted by the relaxation. If

$$\text{conv}(\mathcal{F}_{IP}) = \mathcal{F}_{LP},$$

then the linear relaxation coincides with the integer hull and is therefore an exact convex description of the feasible integer solutions.

Chapter 2

Algorithmic Approaches

In this chapter we explore different methodologies that have been used to show certain properties of the graph root and the graph square. For many restricted cases which are **NP**, a polynomial time reduction is given and in cases which are polynomial, a polynomial time algorithm is given.

2.1 NP-Completeness Proof

We now establish the computational complexity of the graph square problem as was established by Motwani and Sudan in their paper [MS94]. The decision version of the problem asks whether a given graph admits a graph root whose square is the given graph.

Problem 2.1.1 (SQUARE).

Instance: A finite, simple, undirected graph $G = (V, E)$.

Question: Does there exist a finite, simple, undirected graph H such that

$$G = H^2?$$

To show that Problem (2.1.1) is **NP**-complete, it is necessary to show that it is in **NP** and that it is **NP**-hard by reducing not-all-equal SAT (NAESAT) to the problem.

Theorem 1 (NP). *Problem (2.1.1) is in NP.*

NP Proof. Given a candidate square root H , one can verify in polynomial time whether $G = H^2$ by computing the distances between pairs of vertices in H , or equivalently by constructing the square H^2 and comparing its edge set with that of G . Hence, H provides a polynomially verifiable certificate for a **Yes**-instance of SQUARE and therefore

$$\text{SQUARE} \in \text{NP}.$$

□

Problem 2.1.2 (NAESAT).

Instance: A 3-CNF formula

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

over variables $x_1, \dots, x_n$.

Question: Does there exist an assignment of TRUE and FALSE to the variables such that every clause contains at least one TRUE literal and at least one FALSE literal?

Thus, an assignment satisfies the not-all-equal condition if, for every clause, its literals are not all assigned the same truth value. The problem NAESAT is known to be **NP**-complete by Schaefer's dichotomy theorem [Sch78].

2.1.1 NP-Hard proof

SQUARE is **NP**-hard by giving a polynomial-time reduction from **NP**-complete NAESAT and then show that G_ϕ has a graph root if and only if $\phi \in \text{NAESAT}$.

Polynomial Reduction. Let $\phi = C_1 \wedge C_2 \wedge \dots \wedge C_m$ be an instance of NAESAT, where ϕ is a 3-CNF formula on variables $x_1, \dots, x_n$. Over variables $x_1, \dots, x_n$, construct a graph $G_\phi = (V_\phi, E_\phi)$ in polynomial time such that

$$\phi \in \text{NAESAT} \iff G_\phi \text{ has a square root.}$$

The vertex set V_ϕ consists of

- (i) Two constant vertices TRUE and FALSE, plus a vertex x and three tail vertices t_1, t_2, t_3 for x .
- (ii) For each of the $2n$ literals $\ell_1, \dots, \ell_{2n}$, where $\ell_{i+n} = \overline{\ell_i}$, a literal vertex L_i .
- (iii) For every pair of literals ℓ_i, ℓ_k that are not complements of each other, a literal-pair vertex A_{ik} .
- (iv) For each clause $C_j = \{\ell_{j_1}, \ell_{j_2}, \ell_{j_3}\}$, a clause vertex C_j together with three tail vertices C_j^1, C_j^2, C_j^3 .

The edge set E_ϕ defined as

- i) **Tail of x .** The tail of x consists of the edges

$$t_3t_2, \quad t_3t_1, \quad t_2t_1, \quad t_2x, \quad t_1x, \quad t_1\text{TRUE}, \quad t_1\text{FALSE}.$$

- ii) **Tails of the clause vertices.** For every clause C_j , introduce the edges

$$C_j^3C_j^2, \quad C_j^3C_j^1, \quad C_j^2C_j^1, \quad C_j^2C_j, \quad C_j^1C_j, \quad C_j^1L_{j_1}, \quad C_j^1L_{j_2}, \quad C_j^1L_{j_3},$$

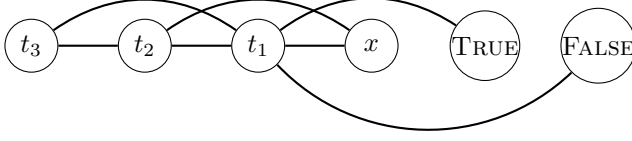
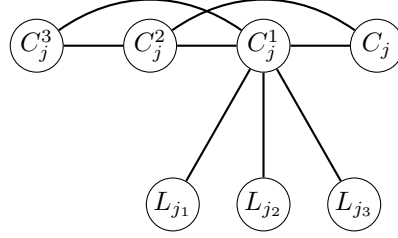
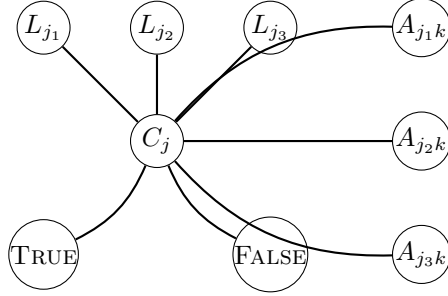
where $C_j = \{\ell_{j_1}, \ell_{j_2}, \ell_{j_3}\}$.

- iii) **Edges incident to clause vertices.** For every clause C_j and every literal $\ell_i \in C_j$, introduce the edge C_jL_i . Also introduce $C_j\text{TRUE}$ and $C_j\text{FALSE}$. Finally, for every literal ℓ_k that is not the complement of a literal in C_j , introduce the edge C_jA_{ik} .
- iv) **Edges among the remaining vertices.** All vertices in

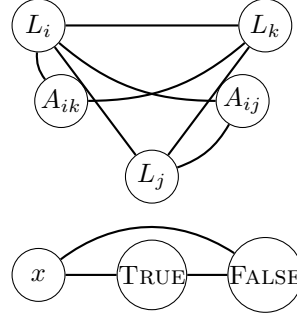
$$\{L_i : 1 \leq i \leq 2n\} \cup \{A_{ik} : \ell_i \neq \overline{\ell_k}\} \cup \{\text{TRUE}, \text{FALSE}, x\}$$

are adjacent to one another, except for pairs of complementary literals. Thus, $L_iL_k \in E_\phi$ whenever ℓ_i and ℓ_k are not complementary, while $L_iL_k \notin E_\phi$ whenever ℓ_i and ℓ_k are complementary.

□

(i) Tail of x (ii) Tail of C_j (iii) Edges incident to C_j 

(iv) Remaining vertices



$\{L_i\} \cup \{A_{ik}\} \cup \{\text{TRUE}, \text{FALSE}, x\}$
is complete, except for complementary literal pairs

The construction is clearly polynomial in $n + m$ and depends only on the formula ϕ . Therefore can be carried out without knowing whether ϕ has a not-all-equal satisfying assignment. Moreover, the number of vertices and edges introduced is polynomial in the size of ϕ .

The following lemma, which is the central combinatorial tool for proving the correctness of the reduction.

Lemma 1 (Tail Lemma). *Let G be a connected graph containing four distinct vertices a, b, c, d such that*

$$N_G(a) = \{b, c\}, \quad N_G(b) = \{a, c, d\}, \quad cd \in E_G,$$

and suppose c has a neighbour in G outside $\{a, b, c, d\}$. If $H^2 = G$, then

$$N_H(d) \setminus \{a, b, c, d\} = N_G(c) \setminus \{a, b, c, d\}.$$

Proof. Since $N_G(a) = \{b, c\}$, every neighbour of a in H is contained in $\{b, c\}$.

First show that $N_H(a) = \{b\}$. Suppose, for contradiction, that $N_H(a) = \{c\}$. Then $ac \in E_H$. Because $ab \in E_G = H^2$, there is a path of length at most two from a to b in H - as a has only the neighbour c , this forces $bc \in E_H$.

Now $cd \in E_G$, so c and d are at distance at most two in H . As $bc \in E_H$, the only possible P_2 from c to d uses b , and hence $bd \in E_H$.

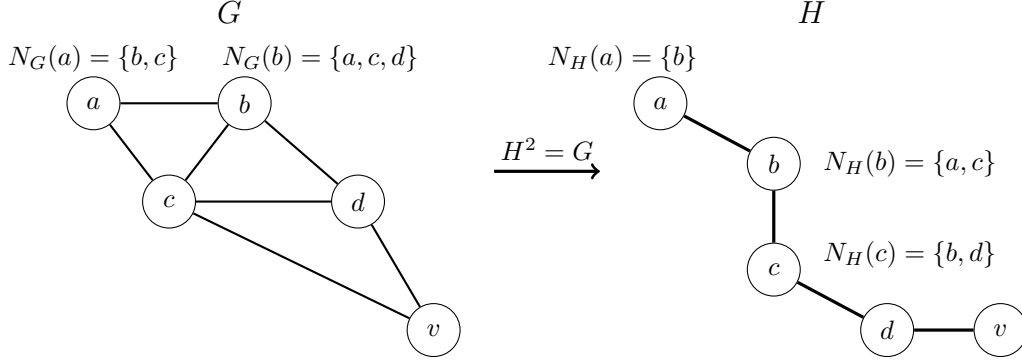


Figure 2.1: The forced local structure in the Tail Lemma. The square root H contains the path $a - b - c - d$. Consequently, $ac, bd \in E_G$ as distance-two edges. For an external neighbour v of c , the Tail Lemma gives $dv \in E_H$, and hence $dv \in E_G$.

Let $v \notin \{a, b, c, d\}$ be the neighbour of c in G . Since $cv \in E_G$, there is a path of length at most two from c to v in H . This path cannot go through a , otherwise $av \in E_G$, contradicting $N_G(a) = \{b, c\}$. Nor can it go through b , since that would give $bv \in E_G$, contradicting $N_G(b) = \{a, c, d\}$. Therefore the path must be $c - d - v$, giving $dv \in E_H$.

Since $bd, dv \in E_H$, have a path $b - d - v$ of length two in H , which implies $bv \in E_G$. This again contradicts $N_G(b) = \{a, c, d\}$. The contradiction eliminates the possibility $N_H(a) = \{c\}$, and consequently

$$N_H(a) = \{b\}. \quad (2.1)$$

Now determine the neighbourhood of c . Since $ab \in E_H$ as $N_H(a) = \{b\}$, and $ac \in E_G$, the edge ac in $G = H^2$ must be realized by the path $a - b - c$. This gives that $bc \in E_H$. If b had a neighbour $x \notin \{a, c\}$ in H , then $a - b - x$ would be a P_2 , forcing $ax \in E_G$, which contradicts $N_H(a) = \{b\}$. Thus

$$N_H(b) = \{a, c\}. \quad (2.2)$$

Since $bd \in E_G$ and $N_H(b) = \{a, c\}$, the P_2 from b to d in H must be $b - c - d$ - it cannot use a , as $N_H(a) = \{b\}$. Therefore $cd \in E_H$.

If c had a neighbour $y \notin \{b, d\}$ in H , then $b - c - y$ would give $by \in E_G$, contradicting Equation (2.2). Hence

$$N_H(c) = \{b, d\}. \quad (2.3)$$

It remains to relate the external neighbours of c in G to those of d in H . Take any $v \notin \{a, b, c, d\}$. If $cv \in E_G$, then there is a path of length at most two from c to v in H . By Equation (2.3), such a path must use b or d as the intermediate vertex. It cannot use b , since that would imply $bv \in E_G$, contradicting $N_G(b) = \{a, c, d\}$. Hence the path is $c - d - v$, so $dv \in E_H$. Conversely, if $dv \in E_H$, then $c - d - v$ is a P_2 in H since $cd \in E_H$, giving $cv \in E_G$. Have therefore established, for every $v \notin \{a, b, c, d\}$,

$$v \in N_G(c) \text{ if and only if } v \in N_H(d).$$

The desired equality follows immediately. $\square$

Lemma 2. *If ϕ is a **Yes-instance** of NAESAT, then G_ϕ has a graph root*

Proof Sketch. Suppose that ϕ has an assignment satisfying not-all-equal. Construct a graph root $H = (V_\phi, E_H)$ on the same vertex set as G_ϕ .

The edges of H consist of the edges of the tails, together with edges connecting x to TRUE and FALSE. For each clause vertex C_j , connect C_j to the three literal vertices corresponding to the literals contained in the clause, together with the edges required by its tail. Each literal vertex is connected to TRUE if its corresponding literal is assigned TRUE and to FALSE if its corresponding literal is assigned FALSE. Finally, each literal vertex is connected to the appropriate literal-pair vertices, and each literal-pair vertex is connected to both TRUE and FALSE.

Because the assignment is not-all-equal, every clause contains at least one TRUE literal and at least one FALSE literal. Consequently, the paths in H of length two produce precisely the edges of G_ϕ . In particular, complementary literals do not have a common neighbour in H , whereas every pair of non-complementary literals has a common neighbour through its corresponding literal-pair vertex. The tail construction similarly ensures that the required tail edges are obtained in H^2 .

Hence, $G_\phi = H^2$, and therefore G_ϕ has a graph root. $\square$

Lemma 3. *If G_ϕ has a graph root, then ϕ is a **Yes-instance** of NAESAT*

Proof Sketch. Suppose that G_ϕ has a graph root H , so that $G_\phi = H^2$.

Apply Lemma 1 to the tail of x , with $(a, b, c, d) = (t_3, t_2, t_1, x)$. Since t_1 has the external neighbours TRUE and FALSE in G_ϕ , the lemma gives

$$N_H(x) \setminus \{t_3, t_2, t_1, x\} = \{\text{TRUE}, \text{FALSE}\}.$$

Thus, apart from its own tail, x is adjacent in H precisely to TRUE and FALSE.

Similarly, for each clause C_j , apply Lemma 1 to its tail with $(a, b, c, d) = (C_j^3, C_j^2, C_j^1, C_j)$. Since C_j^1 has the external neighbours $L_{j_1}, L_{j_2}, L_{j_3}$ in G_ϕ , the lemma gives

$$N_H(C_j) \setminus \{C_j^3, C_j^2, C_j^1, C_j\} = \{L_{j_1}, L_{j_2}, L_{j_3}\}.$$

Hence, apart from its own tail, C_j is adjacent in H precisely to the three literal vertices occurring in C_j . Since every literal vertex L_i is adjacent to x in $G_\phi = H^2$, each L_i must be adjacent in H to at least one of TRUE and FALSE.

On the other hand, complementary literal vertices are non-adjacent in G_ϕ . Therefore they cannot have a common neighbour in H , and hence a literal and its complement cannot both be adjacent to the same one of TRUE and FALSE. It follows that every literal vertex is adjacent to exactly one of TRUE and FALSE, and complementary literals are adjacent to opposite constant vertices.

We may therefore define an assignment by declaring a literal ℓ_i to be TRUE precisely when $L_i \text{TRUE} \in E_H$. Consider any clause $C_j = \{\ell_{j_1}, \ell_{j_2}, \ell_{j_3}\}$. By construction, C_j is adjacent in G_ϕ to both TRUE and FALSE. Since, apart from its own tail, C_j is adjacent in H only to $L_{j_1}, L_{j_2}, L_{j_3}$, these three literal vertices must include one adjacent to TRUE and one adjacent to FALSE.

Thus every clause contains at least one TRUE literal and at least one FALSE literal. Therefore the assignment is a not-all-equal satisfying assignment of ϕ , and hence

$$\phi \in \text{NAESAT}.$$

$\square$

2.2 Polynomial Time Algorithm

Ross and Harary give a necessary and sufficient characterization of graphs that are squares of trees, together with an algorithm for constructing the corresponding tree square root. The characterization is based on the maximal cliques of the square and the way in which these cliques intersect. We

first establish the structural properties of the square of a tree that underlie the characterization, before describing the root finding algorithm.

Theorem 2 (Structural Characterization of Squares of Trees). *Let G be a finite, simple, undirected graph. Then $G = T^2$ for some tree T if and only if either $G = K_p$ is complete, or the maximal cliques $C_1, \dots, C_n$ of G satisfy the following conditions:*

- (i) *G is connected and the closed neighbourhood of every vertex of T forms a clique in G .*
- (ii) *If two maximal cliques have exactly one common vertex v , then there is another maximal clique containing v and one additional vertex from each of the two cliques.*
- (iii) *Each maximal clique C_i can be associated with a unique multicliqual vertex b_i . The number of multicliqual vertices contained in C_i is equal to the number of maximal cliques containing b_i .*
- (iv) *For any two maximal cliques C_i and C_j , $|C_i \cap C_j| \leq 2$, that is their intersection contains at most two points*
- (v) *The number of pairs of maximal cliques whose intersection contains two vertices is $n - 1$.*

To establish this characterization, we first consider several properties of the square of a tree.

Lemma 4. *G is complete if and only if T is a star with depth one*

Proof. Suppose that T has a vertex v that is adjacent to every other vertex. Then every pair of vertices in T is at distance at most two, since any two vertices are either adjacent or have v as a common neighbour. Hence $G = T^2$ is complete.

Conversely, suppose that $G = T^2$ is complete. Then every pair of vertices in T has distance at most two, so $\text{diam}(T) \leq 2$. Since T is a tree, this implies that there is a vertex v adjacent to every other vertex. Thus, choosing v as the root gives a rooted tree T of depth one. $\square$

Lemma 5. *G is connected and the closed neighbourhood of every vertex of T forms a clique in G .*

Proof. Let $v \in V_T$. Every pair of vertices in the closed neighbourhood $N_T[v]$ are either adjacent in T or have v as a common neighbour and thus has distance at most two in T . Hence $N_T[v]$ forms a clique in $G = T^2$. Furthermore, every edge of T is also an edge of G , so T is a spanning subgraph of G . Since T is connected, this implies G is also connected. $\square$

Lemma 6. *Let $G = T^2$ and suppose that G is not complete. Then the maximal cliques of G are precisely the closed neighbourhoods of the non-leaf vertices of T .*

Proof. Let v be a non-leaf vertex of T . Since v has at least two neighbours, every pair of vertices in $N_T(v)$ is at distance two through v . Hence $C(v) = N_T[v]$ is a clique in G . Now consider a vertex $u \notin N_T[v]$. The unique path from u to v in T has length at least two. If this path has length greater than two, then u is not adjacent in G to v . If the path has length two, then u can only be adjacent in G to the vertices along the corresponding part of the path. In either case, u cannot be added to $N_T[v]$ while maintaining a clique. Therefore $N_T[v]$ is a maximal clique.

Conversely, let C be a maximal clique of G . Since $G = T^2$, vertices in C are pairwise at distance at most two in T . The vertices of such a clique must therefore be contained in the closed neighbourhood of a common non-leaf vertex v of T . Thus $C \subseteq N_T[v]$. Since $N_T[v]$ is itself a clique and C is maximal, this gives $C = N_T[v]$. Hence every maximal clique of G is the closed neighbourhood of a non-leaf vertex of T . $\square$

Lemma 7. *If $G = T^2$ is not complete, then a vertex of G belongs to exactly one maximal clique if and only if it is a leaf of T .*

Proof. Let v be a leaf of T , and let u be its unique neighbour. By Lemma 6, every maximal clique corresponds to the closed neighbourhood of a non-leaf vertex. Since v has only one neighbour, the only maximal clique containing v is $N_T[u]$. Thus v belongs to exactly one maximal clique.

Conversely, suppose that v is not a leaf. Then v has at least two neighbours in T . By Lemma 6, $N_T[v]$ is a maximal clique containing v . Since G is not complete, Lemma 4 implies that T is not a star. Hence at least one neighbour u of v is a non-leaf vertex. The closed neighbourhood $N_T[u]$ is therefore another maximal clique containing v . Thus v belongs to at least two maximal cliques. Therefore, the vertices belonging to exactly one maximal clique are exactly the leaves of T . $\square$

Lemma 8. *Let C_i and C_j be maximal cliques corresponding to non-leaf vertices v_i and v_j of T . Then $|C_i \cap C_j| = 2$ if and only if $v_i v_j \in E_T$.*

Proof. Suppose that v_i and v_j are adjacent in T . Then both v_i and v_j belong to $N_T[v_i] \cap N_T[v_j]$. If there were another vertex u in this intersection, then u would be adjacent to both v_i and v_j in T . Together with the edge $v_i v_j$, this would create a cycle in T , which is impossible. Therefore, $|N_T[v_i] \cap N_T[v_j]| = 2$. By Lemma 6, these closed neighbourhoods are the corresponding maximal cliques, so $|C_i \cap C_j| = 2$.

Conversely, suppose that $|C_i \cap C_j| = 2$. Since v_i and v_j belong to their respective closed neighbourhoods, the intersection implies that v_i and v_j are either adjacent or have a common neighbour in T . If they were not adjacent, there would be a vertex u adjacent to both v_i and v_j . Then $v_i, v_j, u \in C_i \cap C_j$, which would give an intersection of at least three vertices, contradicting $|C_i \cap C_j| = 2$. Therefore, $v_i v_j \in E_T$. $\square$

Lemma 9. *G is a connected graph where the removal of any edge does not result in a disconnected graph*

Proof. Let $v \in V_G$. Consider two vertices $x, y \in V_G \setminus \{v\}$ that lie in different components of $T - v$. Since T is a tree, the vertices in each component that are closest to v are neighbours of v in T . These vertices are therefore adjacent in $G = T^2$. Thus the different components of $T - v$ are connected through vertices that remain in $G - v$.

Consequently, removing v from G does not disconnect the graph. Hence G has no cut vertex. $\square$

Combining gives the following proof of Theorem (2)

Proof. Suppose first that $G = T^2$ for some tree T . If G is complete, then Lemma 4 shows that T is a star.

Now suppose that G is not complete. By Lemma 5, G is connected and every closed neighbourhood in T forms a clique in G , giving (i). By Lemma 6, each maximal clique corresponds to a non-leaf vertex of T . The way in which these closed neighbourhoods overlap gives (ii) and (iii). By Lemma 8, two maximal cliques intersect in two vertices exactly when their corresponding non-leaf vertices are adjacent in T . Since T contains no cycles, two such closed neighbourhoods cannot have more than two vertices in common. This gives (iv). Finally, the non-leaf vertices of T , together with the edges between them, form a tree. If there are n such vertices, there are exactly $n - 1$ edges between them. By Lemma 8, each such edge corresponds to a pair of maximal cliques with an intersection containing exactly two vertices. Hence (v) holds.

Conversely, suppose that G is not complete and that its maximal cliques $C_1, \dots, C_n$ satisfy (i)–(v). By (iii), associate each maximal clique C_i with a unique vertex b_i that belongs to multiple

maximal cliques. These vertices will form the internal vertices of the tree root. Construct a graph T with vertex set $V_T = \{b_1, \dots, b_n\}$. For every pair of maximal cliques satisfying $|C_i \cap C_j| = 2$, add the edge $b_i b_j$ to T . By (v), T contains exactly $n - 1$ edges, and the intersection structure given by the characterization ensures that these edges connect all vertices without forming a cycle. Thus T is a tree. It remains to add the uniclqual vertices. By (iii), these vertices correspond to leaves of the root. For each clique C_i , attach all vertices of C_i that occur in no other maximal clique as leaves adjacent to b_i . The resulting graph is a tree T . Each maximal clique C_i is represented by the closed neighbourhood of b_i in T , while the remaining vertices are attached as leaves to their corresponding internal vertices. Therefore every edge of G is represented by two vertices at distance at most two in T , and no additional edges are introduced. Hence $T^2 = G$ and G is the square of a tree. $\square$

The proof for Theorem (2) gives the following algorithm. Given an inputted tree square, by identifying all maximal cliques, finding all points which are in multiple cliques and those in only one clique, and constructing the tree T by appropriately connecting multicliqual points together as internal vertices and uniclqual points as the tree leaves.

Algorithm 1 Ross-Harary Tree Construction

1: **Input:** A graph G where $G = T^2$

2: **Output:** A tree T

Step 1

3: **if** $G = K_p$ is complete **then**

4: **Return** tree T with depth 1 where root can be any vertex

5: **else**

6: **Continue**

7: **end if**

Step 2

8: Identify all maximal cliques $C_1, \dots, C_n$ of G using an algorithm such as Bron-Kerbosch where $n > 1$ from characterization (i)

Step 3

9: Let $b_1, \dots, b_n$ be multicliqual points corresponding to $C_1, \dots, C_n$ formed from characterization (iii)

10: Construct a graph T with vertex set $\{b_1, \dots, b_n\}$

11: **for** each pair (C_i, C_j) satisfying $|C_i \cap C_j| = 2$ **do**

12: Add edge between b_i and b_j in S

13: **end for**

14: By characterization (v), T is a tree

Step 4

15: **for** each clique C_i **do**

16: Let n_i be the number of uniclqual points in C_i

17: Attach n_i leaf nodes to b_i in T

18: **end for**

19: The resulting tree T has the same number of nodes as G and satisfies $T^2 = G$

Theorem 3 (Tree Root Algorithm). *Given a graph $G = T^2$, Algorithm (1) returns a tree T such that $T^2 = G$. Furthermore, this tree is unique up to isomorphism.*

Proof. If G is complete, Step 1 returns a star. By Lemma 4, this is a tree square root of G . Otherwise, Lemma 6 shows that each maximal clique corresponds to a non-leaf vertex of the root. Thus the vertices $b_1, \dots, b_n$ identified in Step 3 determine the internal vertices of T . By Lemma 8, two of these internal vertices are adjacent precisely when their corresponding maximal cliques have an intersection of size two. Therefore Step 3 recovers all edges between the internal vertices. The remaining vertices belong to exactly one maximal clique and, by Lemma 7, correspond to leaves. Step 4 therefore places each of these vertices at its unique position in the tree. Hence the algorithm reconstructs a tree T satisfying $T^2 = G$. Since the internal vertices, their adjacencies, and the attachment of every leaf are all determined by the maximal cliques of G , the resulting tree is unique up to isomorphism. $\square$

2.3 Integer Program

The two methods so far have primarily focused on the definition of a square of a graph to be paths of length smaller than two defined in Definition (1.3.38). Using adjacency matrices, this definition is equivalent expressed as

$$A_G = (A_H)^2 \vee A_H. \quad (2.4)$$

where $\vee$ is applied entrywise to the matrices. This concise interpretation with the logical operator lends itself easily to a SAT problem whereby one could introduce Boolean variables for the edges of H and encode $(A_H)_{ij}^2 = ((A_H)_{i1} \wedge (A_H)_{1j}) \vee ((A_H)_{i2} \wedge (A_H)_{2j}) \vee \dots \vee ((A_H)_{in} \wedge (A_H)_{nj})$. Thus, for each pair of distinct vertices i, j , the condition that ij is an edge of G can be expressed as

$$g_{ij} \text{ if and only if } \left(\bigwedge_k (x_{ik} \wedge x_{kj}) \vee x_{ij} \right). \quad (2.5)$$

The resulting Boolean formula can be converted into a standard SAT instance using auxiliary variables and a polynomial number of clauses. Hence, the logical formulation provides a polynomial-size SAT representation of the graph-square condition.

However, constructing a polynomial-size SAT instance does not imply that the underlying problem can be solved in polynomial time. A general SAT instance may require exponential time to solve, since SAT is NP -complete. Therefore, to obtain a polynomial-time algorithm through this approach, one would additionally need to show that the SAT instances arising from Equation (2.5) belong to a tractable subclass of SAT, such as 2-SAT, or otherwise provide a polynomial-time algorithm for solving these particular instances. The formulation in Equation (2.5) evidently does not belong to 2-SAT, as each constraint involves a disjunction over all possible common neighbours k and therefore contains terms involving more than two Boolean variables. More generally, one would suspect from the results in 2.1 that there is no immediate polynomial-time algorithm for solving this instance and any restricted version of the graph root finding problem would necessitate significant machinery.

Instead, we aim to situate the problem completely within a polynomial-time framework by considering another equivalent representation of Equation (2.4) and its equivalent formulation in Definition (1.3.38).

Lemma 10. *The graph G defined by Definition (1.3.39) is equivalent to the graph square H^2 defined in Definition (1.3.38)*

$$ij \in E_{H^2} \text{ if and only if } 1 \leq d_H(i, j) \leq 2. \quad (2.6)$$

Proof. For $i \neq j$, since D_H is diagonal, have $(D_H)_{ij} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ij} = (A_H)_{ij}^2 + (A_H)_{ij}.$$

Suppose $ij \in E_G$ such that $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$. This implies that at least one of $(A_H)_{ij} = 1$ or $(A_H)_{ij}^2 \geq 1$ must be true. If $(A_H)_{ij} = 1$, then $d_H(i, j) = 1$. If $(A_H)_{ij}^2 \geq 1$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $d_H(i, j) \leq 2$. Therefore $1 \leq d_H(i, j) \leq 2$, and thus $ij \in E_{H^2}$.

Suppose $ij \in E_{H^2}$ such that $1 \leq d_H(i, j) \leq 2$. D_H is diagonal such that $(D_H)_{ij} = 0$. If $d_H(i, j) = 1$, then $(A_H)_{ij} = 1$. If $d_H(i, j) = 2$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $(A_H)_{ij}^2 \geq 1$. In either case, $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$, so $(A_G)_{ij} = 1$, and hence $ij \in E_G$.

For $i = j$, since all graphs are simple, have $(E)_{ii} = 0$, $(A)_{ii} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ii} = (A_H)_{ii}^2 - (D_H)_{ii}.$$

$(A_H)_{ii}^2 - (D_H)_{ii} = 0$ is always true since the degree matrix of H always has diagonal values $(D_H)_{ii} = \sum_k x_{ik}$ and each edge from vertex i to its neighbour in H will always create a loop to itself in the square such that $(A_H)_{ii}^2 = \sum_{k=1} x_{ik} x_{ki} = \sum_{k=1} x_{ik}^2$ by the symmetry of adjacency matrix A_H . Since $x_{ik} \in \{0, 1\}$, the square of the boolean is identical to the boolean x_{ik} such that $(A_H)_{ii}^2 = \sum_k x_{ik} = (D_H)_{ii}$ and $(A_G)_{ii} = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H)$ as required.

Combining both gives that $E_G = E_{H^2}$, and so $G = H^2$. $\square$

The formulation of the adjacency matrix of the graph square in Equation (1.2) is very powerful. However, the presence of the indicator function introduces a nonlinearity that prevents the problem from being expressed as a purely quadratic form, thereby limiting the direct applicability of existing results from the quadratic optimization literature.

Lemma 11. *If there are no K_3 triangle subgraphs or any C_4 square subgraphs in the graph root such that H has girth of length 5, then Equation (1.2) can be simplified to become*

$$A_G = ((A_H)^2 - D_H + A_H). \quad (2.7)$$

Proof. Since the graph root has girth 5 there are no cycles of length 3 or 4 in H . Thus for every pair of vertices $i \neq j \in V$ there is either no cycle that connects i, j or there is a cycle of at least length 5 between them. In all cases $(D_H)_{ij} = 0$. If there is no cycle, then there is at most one path between i, j

- i) If the path is of length 1, then $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.
- ii) If the path is of length 2, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.
- iii) If the path is of length ≥ 3 or a path does not exist, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If there is a cycle of length ≥ 5 between i, j such that there are two paths between i, j

- i) If one path is of length 1, then the other is of length ≥ 4 and hence $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.
- ii) If one path is of length 2, then the other is of length ≥ 3 and hence $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.
- iii) If the paths are both of length ≥ 3 , then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If $i = j$, then $(A_H)_{ii} = 0$, $(D_H)_{ii} = \deg_H(i)$, and $(A_H)_{ii}^2 = \deg_H(i)$. Thus,

$$(A_H)_{ii}^2 - (D_H)_{ii} + (A_H)_{ii} = \deg_H(i) - \deg_H(i) + 0 = 0 = (A_G)_{ii}.$$

□

We will not focus on this particular restricted class and analyze the integer program generated by Equation (2.7), but rather aim for a unified method for solving the problem in general. Nevertheless this result gives strong intuition into the results established by Farzad, Lau, Le, and Tuy in [Far+09] in constructing a polynomial time algorithm for recognizing graph roots where H has girth at least 6.

2.3.1 Formulation

Let $x_{ij} \in \{0, 1\}$ be the decision variables of A_H , let $y_{ikj} \in \{0, 1\}$ be the decision variable for $x_{ik}x_{kj}$ in $(A_H)_{ij}^2$, and let $z_{ij} \in \mathbb{Z}_{\geq 0}$ represent the argument inside the indicator function.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, \quad A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix} \text{ such that}$$

$$(A_H)^2 = \begin{bmatrix} \sum_{k=1}^n x_{1k}x_{k1} & \cdots & \sum_{k=1}^n x_{1k}x_{kj} & \cdots & \sum_{k=1}^n x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{ik}x_{k1} & \cdots & \sum_{k=1}^n x_{ik}x_{kj} & \cdots & \sum_{k=1}^n x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{nk}x_{k1} & \cdots & \sum_{k=1}^n x_{nk}x_{kj} & \cdots & \sum_{k=1}^n x_{nk}x_{kn} \end{bmatrix}.$$

Then with known constants $g_{ij} \in \{0, 1\}$ from A_G , can reformulating (1.2) as an integer program

by equating entries of each matrix to give the following linear system of n^2 equations.

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{k1} - (D_H)_{11} + x_{11}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\right) \end{bmatrix}. \quad (2.8)$$

The matrix D_H is diagonal can be represented as $(D_H)_{ii} = \sum_k x_{ik}$ where $(D_H)_{ij} = 0$ for $i \neq j$.

The known z_{ij} represent the terms inside the indicator function such that

$$g_{ij} = 1 \text{ if and only if } z_{ij} \geq 1 \text{ and } g_{ij} = 0 \text{ if and only if } z_{ij} = 0.$$

Let $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant. In this case, M is bounded by the number of edges such that it can be at most $n(n-1) = O(n^2)$. The above condition can be modelled using a standard big- M formulation, and the resulting integer program is given by

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (2.9)$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (2.10)$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (2.11)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (2.12)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (2.13)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (2.14)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (2.15)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (2.16)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (2.17)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (2.18)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (2.19)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (2.20)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (2.21)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (2.22)$$

The construction of the integer program on the arbitrary graph G using Equations (2.9)–(2.22) is denoted as $\mathcal{IP}(G)$.

Additional constraints arising from symmetry and diagonality are added to the integer program which the adjacency matrix x must satisfy. Although analogous constraints could also be imposed on the auxiliary variables z they are omitted due to redundancy and potentially introduces artificial degeneracy in the linear program.

2.3.2 Validity

Theorem 4. *Let G be a graph and consider the integer program whose variables $x_{ij} \in \{0, 1\}$ encode a graph H with adjacency matrix $A_H = (x_{ij})$. Then G has a square root H if and only if the integer program has a feasible solution x such that the corresponding graph H satisfies $H^2 = G$.*

Proof. Suppose G has a square root H and let A_H be the adjacency matrix of H . $x_{ij} = (A_H)_{ij}$. Construct the variables x_{ij}, y_{ikj}, z_{ij} and the constant g_{ij} as such

i) Let $x_{ij} = 1$ if $ij \in E_H$ and $= 0$ otherwise.

ii) Let $y_{ikj} = x_{ik}x_{kj}$ for all i, k, j .

iii) Let $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}$ for all i, j .

- iv) Let $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} = \sum_{k=1}^n x_{ik}^2 - \sum_{k=1}^n x_{ik} = 0$ since $x_{ij} \in \{0, 1\} \implies x_{ik}^2 = x_{ik}$.
- v) Let $g_{ij} = 1$ if $ij \in E_G$ and $= 0$ otherwise.

Verifying constraints of the integer program for the variables

- i) As $x_{ij} \in \{0, 1\} \implies y_{ikj} \in \{0, 1\}$, Equations (2.9)-(2.11) and Equations (2.19)-(2.20) are satisfied.
- ii) As H is a simple directed graph, its adjacency matrix is hallow and symmetric such that Equations (2.21) -(2.22) are satisfied.
- iii) As $y_{ikj} = x_{ik}x_{kj}$ and all variables are binary, Equations (2.16)- (2.18) are satisfied.
- iv) Equation (2.13) by definition is satisfied.
- v) As $z_{ii} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{k=1}^n x_{ik}$, Equation (2.12) is satisfied.
- vi) As $z_{ij} \geq 1$ if and only if $x_{ij} = 1$ or there exists k such that $x_{ik} = x_{kj} = 1$ and as $H^2 = G$, have $g_{ij} = 1$ if and only if $1 \leq d_H(i, j) \leq 2$. Thus $g_{ij} = 1$ if and only if $z_{ij} \geq 1$ so the indicator constraints Equations (2.14)-(2.15) hold.

All constraints of the integer program are satisfied by the constructed variables. Hence, the integer program is feasible.

Suppose the integer program on graph G has a feasible solution x , and let x_{ij} define a graph H by

$$ij \in E_H \text{ if and only if } x_{ij} = 1.$$

As x_{ij} satisfies the constraints of the integer program,

- i) Equation (2.9) and Equation (2.19) gives $x_{ij} \in \{0, 1\}$ for all i, j ,
- ii) Equation (2.21) and Equation (2.22) give $x_{ij} = x_{ji}$ and $x_{ii} = 0$,

such that x_{ij} defines a hollow, symmetric matrix with all elements being either 0 or 1 which is a valid adjacency matrix of a graph. Examining the auxiliary variables y_{ikj} and z_{ij} ,

- i) Equations (2.16)-(2.18) give the strict equality $y_{ikj} = x_{ik}x_{kj}$ for all i, j, k ,
- ii) Equation (2.13) gives $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij} = \sum_{k=1}^n x_{ik}x_{kj} + x_{ij}$ for all $i \neq j$,
- iii) Equation (2.12) gives $z_{ii} = \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{k=1}^n x_{ik} = 0$ for all i since

$$\sum_{k=1}^n x_{ik}x_{ki} = \sum_{k=1}^n x_{ik}^2 = \sum_{k=1}^n x_{ik},$$

by symmetry and since $x_{ij} \in \{0, 1\}$.

The indicator constraints Equations (2.14)-(2.15) thus give for $i \neq j$

$$g_{ij} = 1 \text{ if and only if } \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} \geq 1 \text{ and } g_{ij} = 0 \text{ if and only if } \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} = 0.$$

Since $x_{ij} \in \{0, 1\}$, this is equivalent to

$$g_{ij} = 1 \text{ if and only if } (x_{ij} = 1) \vee \left(\text{there exists } k \text{ such that } x_{ik} = 1 \wedge x_{kj} = 1 \right) \text{ and } g_{ij} = 0 \text{ if and only if } x_{ij} = 0.$$

By definition of H this means that $g_{ij} = 1$ if and only if $(ij \in E_H) \vee \left(\text{there exists } k \text{ such that } ik, kj \in E_H \right)$ which is equivalent to

$$g_{ij} = 1 \text{ if and only if } 1 \leq d_H(i, j) \leq 2.$$

Applying Lemma 10 thus gives $ij \in E_G$ if and only if $ij \in E_H^2$ such that $G = H^2$. □

The correctness of the integer program is elucidated by Figure (3).

Chapter 3

Square of A Tree using Linear Programming

3.1 Linear Program Relaxation

The linear relaxation of the integer program defined by Equations (2.9)–(2.22) is constructed in order to generate a feasible weighted solution in polynomial time. It is done with identical constraints, however the decision variables (2.9), (2.10) and (2.11) are replaced with

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (3.1)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (3.2)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (3.3)$$

The construction of the linear program on the arbitrary graph G using Equations (3.1)–(3.3) and Equations (2.12)–(2.22) is denoted as $\mathcal{LP}(G)$.

The results from the linear programming can give *integral solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist such that the results in Figure 4 can be generated.

Definition 3.1.1 (Integral Solutions). *A feasible solution to the linear program is an integral solution if*

$$x_{ij} \in \{0, 1\} \quad \forall i, j \in V.$$

In this case, the variables x_{ij} define a graph root H through

$$E_H = \{(i, j) \in V \times V : x_{ij} = 1\}.$$

Definition 3.1.2 (Fractional Solutions). *A feasible solution to the linear program is a fractional solution if*

$$x_{ij} \in [0, 1] \quad \forall i, j \in V,$$

and there exists at least one pair (i, j) such that $0 < x_{ij} < 1$. In this case, the variables x_{ij} do not directly define a graph root.

In Figure 3.1, solutions which are LP feasible may be integral or fractional

The correctness of the integer program is elucidated by Figure (4).

The linear relaxation potentially breaks equations in the integer program which are critical to the definition.

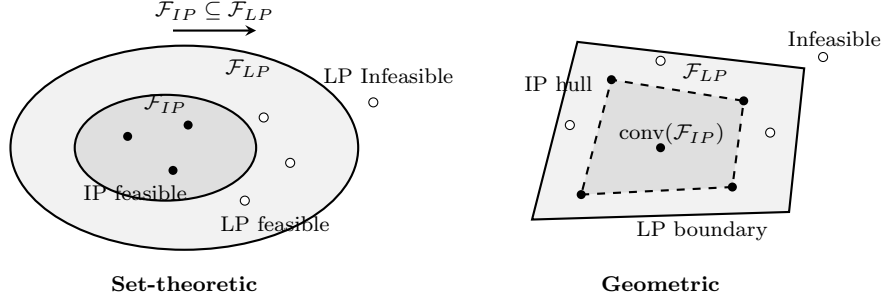


Figure 3.1: Set-theoretic and geometric relationships between an integer program and its linear relaxation.

Consider Equations (2.16)–(2.18) which gives $y_{ikj} = x_{ik}x_{kj}$. When x is binary, this construction of y leads to y being binary as well.

Consider Equation (2.12), where $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} + x_{ii}$ can be simplified to $z_{ii} = \sum_{k=1}^n x_{ik}^2 - \sum_{k=1}^n x_{ik}$ and is equal to 0 when x is binary. In the linear relaxation this strict inequality no longer holds for fractional values of x_{ik} . However, when x_{ik} is fractional $x_{ik}^2 \leq x_{ik}$ such that $z_{ii} = \sum_{k=1}^n x_{ik}^2 - x_{ik} \leq 0$. As $z \in \mathbb{Z}_{\geq 0}$, this gives that $z_{ii} = 0$ as required - no additional modifications are required.

Consider Equation (2.13), where $z_{ij} = \sum_{k=1}^n x_{ik}x_{kj} + x_{ij}$. When $z_{ij} = 0$, the equation follows the standard definition of graph squares such that $x_{ij} = 0$ and for all intermediary k s at least one of x_{ik}, x_{kj} must be equal to 0. However, when $z_{ij} \geq 1$, fractional solutions of x_{ij} and $x_{ik}x_{kj}$ may satisfy the constraints without giving a valid integral path in the root.

Proposition 1. $\mathcal{LP}(G)$

3.2 Min Linear Program

Consider the addition of an objective function maximizing the total weights on all edges to the linear relaxation defined by Equations (2.12)–(3.3) in Section 3.1

$$\min \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (3.4)$$

The construction of the linear program on the arbitrary graph G using Equations (3.1)–(3.3), Equations (2.12)–(2.22) together with Equation (3.4) is denoted as $\min \mathcal{LP}(G)$.

This relaxation however yields more fractional solutions, as illustrated in Figure 6.

Lemma 12 (Fractional Cliques in $\min \mathcal{LP}(G)$). *Let $C \cong K_n$ be a complete subgraph. The minimizing solution of $\min \mathcal{LP}(G)$ assigns the same value to every edge of graph G , such that for $i \neq j$*

$$x_{ij} = \frac{1}{n-1}.$$

In particular, K_3 gives $x_{ij} = \frac{1}{2}$, K_4 gives $x_{ij} = \frac{1}{3}$, and in general K_n gives $x_{ij} = \frac{1}{n-1}$.

Proof. Since $C \cong K_n$ is vertex-transitive, the feasible region and the objective function Equation (3.4) are invariant under every permutation of the vertices of G . Hence, if x is optimal, then averaging x over all permutations of the vertex set V produces another optimal solution $\bar{x}$ with

$$\bar{x}_{ij} = w \quad \text{for all } i \neq j \in V$$

for some constant $w \geq 0$. Consider any vertex $i \in V$, where in the clique every pair (i, j) with $j \neq i$ is present, so that $g_{ij} = 1$. By the indicator constraints Equations (2.14)–(2.15),

$$z_{ij} \geq g_{ij} = 1 \quad \text{for all } j \neq i.$$

Together with Equation (2.13) and the product constraints Equations (2.16)–(2.18), this forces each vertex i to carry at least one unit of total incident x -mass on the clique,

$$\sum_{j \in V(C) \setminus \{i\}} x_{ij} \geq 1.$$

For the symmetric solution $\bar{x}$, this becomes $(n-1)w \geq 1$, hence $w \geq \frac{1}{n-1}$. On the other hand,

the objective function is $\sum_{i < j, i, j \in V(C)} x_{ij} = \binom{n}{2} w$, which is strictly increasing in w . Therefore the

minimum is attained at the smallest feasible value of $w = \frac{1}{n-1}$ which every edge of K_n achieves in the minimizing LP. $\square$

3.3 Max Linear Program

Consider the addition of an objective function maximizing the total weights on all edges to the linear relaxation defined by Equations (2.12)–(3.3) in Section 3.1

$$\max \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \tag{3.5}$$

The construction of the linear program on the arbitrary graph G using Equations (3.1)–(3.3), Equations (2.12)–(2.22) together with Equation (3.5) is denoted as $\max \mathcal{LP}(G)$.

This objective encourages dense solutions and therefore tends to produce relaxations that are more dense and hence more informative than trivial sparse solutions. In particular, maximizing the total edge weight pushes the variables toward values closer to 1, which promotes direct edges and reduces the need for fractional paths of length 2 to satisfy Equation (2.13). As a result, the relaxation is more likely to yield integral solutions, as illustrated in Figure 5.

This choice of objective is also useful when identifying square root solutions beyond those unique up to isomorphism. For example, the complete graph K_3 has four roots: itself and three path graphs of length 2, which are isomorphic to one another. This illustrates that the maximization objective preserves consistency up to isomorphism while still distinguishing structurally meaningful solutions.

3.3.1 Fundamental Properties

Proposition 2. *Consider the graph square G where the edge between vertices a and b is missing, $ab \notin E_G$ such that $g_{ab} = 0$. Then every feasible solution of $\mathcal{LP}(G)$ satisfies $x_{ab} = 0$ and $y_{avb} = 0$ for all vertices $v \in V \setminus \{a, b\}$.*

Proof. Using Equations (2.14)–(2.15) if $g_{ab} = 0$ then $z_{ab} \leq 0$ and $z_{ab} \geq 0$ such that $z_{ab} = 0$. Then Equation (2.13) gives that $0 = \sum_{v=1}^n y_{avb} + x_{ab}$. Since each x, y variable are non-negative, this only holds when each variable is equal to 0 which gives that $x_{ab} = 0$ and $y_{avb} = 0$ for all vertices $v \in V \setminus \{a, b\}$. $\square$

Proposition 3. *Consider the graph square G where the edge between vertices a and b is missing, $ab \notin E_G$ such that $g_{ab} = 0$. Then every feasible solution of $\mathcal{LP}(G)$ satisfies that for all vertices $v \in V \setminus \{a, b\}$, $x_{av} + x_{vb} \leq 1$.*

Proof. Since Proposition (2) gives $y_{avb} = 0$, using Equation (2.15) gives that $0 = y_{avb} \geq x_{av} + x_{vb} - 1$ which after rearranging gives $x_{av} + x_{vb} \leq 1$ as required. $\square$

Proposition 4. *Consider the graph square G where $ab \in E_G$ but it is known that ab cannot be in the root H , that is $x_{ab} = 0$. Then every feasible solution of $\mathcal{LP}(G)$ satisfies*

Proof. $\square$

3.4 Leaf Fixing

There are two important structures in a tree.

Definition 3.4.1 (Pendant). *A pendant subgraph is the induced subgraph on three vertices s, u, v where s is a leaf with unique neighbour u , and u is a stem of degree 2 with neighbours s and v , where v connects the subgraph to the remainder of the tree.*

Definition 3.4.2 (Pendant Fork). *A pendant fork subgraph is the induced subgraph on four vertices s_1, s_2, u, v where s_1, s_2 are leaves with unique neighbour u , and u is an internal vertex of degree ≥ 3 where $\{s_1, s_2, v\} \subseteq N(u)$, where v connects the subgraph to the remainder of the tree.*

Lemma 13. *Every tree contains either a pendant or a pendant fork subgraph.*

Proof. Let T be a finite tree with $|V(T)| \geq 3$, and root T at an arbitrary vertex. Since T is finite, there exists a leaf s of maximal depth where its parent is u . Every child of u is a leaf, since if there was some child w of u that is not a leaf, then w would have a child and hence there would exist a vertex at depth strictly greater than that of s , contradicting the choice of s .

If u is not the root, let v denote its parent such that every neighbour of u other than v is a leaf.

If u has exactly one leaf neighbour s besides v (when u is not the root), then u has exactly two neighbours, namely s and v , hence $\deg_T(u) = 2$. Therefore the induced subgraph on

$$\{s, u, v\}$$

forms a pendant subgraph.

Otherwise u has at least two leaf neighbours s_1, s_2 . Choose another neighbour v of u distinct from s_1, s_2 ; if u is not the root, take its parent, while if u is the root choose any remaining neighbour. Then

$$\{s_1, s_2, v\} \subseteq N(u),$$

and $\deg_T(u) \geq 3$. Hence the induced subgraph on

$$\{s_1, s_2, u, v\}$$

forms a pendant fork subgraph. $\square$

Using this lemma, every tree contains either a pendant or a pendant fork subgraph. Consequently, whenever G admits a tree root, at least one of these two configurations can be identified and used to derive valid fixings for the variables x_{ij} .

3.4.1 Single Triangular Decomposition

For a vertex $s \in V$, define

$$T(s) = \{\{u, v\} \subseteq V \setminus \{s\} \mid (s, u) \in E, (s, v) \in E, (u, v) \in E\}$$

to be the set of K_3 s or triangles in G that contain s . If $|T(s)| = 1$ where $T(s) = \{\{u, v\}\}$ then the vertex s exists in a unique triangle of G . When the neighbours have different degree $\deg_G(u) \neq \deg_G(v)$, then the vertex with smaller degree must be the stem vertex in the square root. This is assigned deterministically for a single triangle s using

$$x_{sa} = x_{as} = x_{ab} = x_{ba} = 1, \quad \text{where } (a, b) = \begin{cases} (u, v), & \deg_G(u) \leq \deg_G(v), \\ (v, u), & \deg_G(u) > \deg_G(v). \end{cases} \quad (3.6)$$

If $\deg_G(u) = \deg_G(v)$, then either u, v can be chosen since they are connected to a clique.

Lemma 14. *Let s, u, v form a triangle in square G where s only lies in a single clique. If $\deg_G(u) = \deg_G(v) = d \geq 2$, then*

$$N_G(u) \setminus \{s\} = N_G(v) \setminus \{s\}.$$

That is, $N_G(u) \cup \{u\}$ and $N_G(v) \cup \{v\}$ induce the same clique structure in G .

Proof. Since s lies in a single clique in G , in any tree root H the vertex s must be a leaf, and exactly one of u or v is the internal vertex adjacent to the remainder of the graph. Assume without loss of generality that u is this internal vertex. Any vertex adjacent to v in G other than s must arise from a path of length at most two through u in H , since v has no other connections in the root besides u and s . Hence

$$N_G(v) \setminus \{s\} \subseteq N_G(u) \setminus \{s\}.$$

Since $\deg_G(u) = \deg_G(v)$, both neighbourhoods have equal cardinality $|N_G(v)| = |N_G(u)|$ such that via the equality criterion for finite sets

$$N_G(v) \setminus \{s\} = N_G(u) \setminus \{s\}.$$

Hence the vertices adjacent to u and v outside of s have identical adjacency structure in G , and therefore $N_G(u) \cup \{u\}$ and $N_G(v) \cup \{v\}$ induce the same clique structure. $\square$

Proposition 5. *Let s, u, v form a triangle in square G where s only lies in a single clique. If $\deg_G(u) = \deg_G(v) = d \geq 2$, then one can choose to assign $x_{su} = x_{uv} = 1$ in the tree root H without loss of generality.*

Proof. Since s lies in a single clique in G , in the root H , s must be a leaf with either u or v being the internal vertex adjacent to the remainder of the graph. Assume without loss of generality that u is the internal vertex.

By Lemma 14, $N_G(u) \setminus \{s\} = N_G(v) \setminus \{s\}$, such that u and v induce identical clique structure in G . Therefore exchanging the roles of u and v preserves the square graph G . Hence choosing u as the internal vertex and assigning

$$x_{su} = x_{uv} = 1,$$

does not alter feasibility, since the alternative choice obtained by exchanging u and v produces an isomorphic tree root with identical square graph. $\square$

It remains to verify that whenever a root H of G exists, the addition of the constraint Equation (3.6) preserves the feasibility of the linear program.

Proposition 6. *Let s, u, v form a triangle in square G where s only lies in a single clique. The linear program with Equation (3.6) never returns a infeasible solution for squares with roots.*

Proof. $\square$

3.4.2 Tetrahedron Decomposition

A natural extension of Section 3.4.1 can be achieved by considering cliques of size greater than three in order to categorize tree structure whereby an internal vertex is connected to more than one leaf. Whilst finding a clique of size $\geq k$ for fixed k is polynomial, with complexity of order $O(n^k)$ by simply checking all tuples of size k , the decision problem of whether a graph G contains a clique of size k , as well as the problem of finding a largest clique, is NP-complete. Thus, it is necessary to only consider cliques of fixed size 4 which have polynomial complexity $O(n^4)$. This however is sufficient for classifying any pendant fork, since any branching vertex with exactly two leaves will generate a K_4 in the square, whilst a branching vertex with more than two leaves will generate multiple K_4 s of which one can be chosen by arbitrarily selecting any two leaves.

Consider vertices $s_1, s_2, u, v \in V$ such that $K_4 = \{s_1, s_2, u, v\}$ induces a K_4 in G and for each leaf s_1, s_2 of u their neighbourhoods are equivalent and give the common neighbourhood $N_L = N_G(s_1) \setminus K_4 = N_G(s_2) \setminus K_4$. If

$$(N_G(s_1) \setminus K_4) = (N_G(s_2) \setminus K_4) \subset (N_G(u) \setminus K_4) \subseteq (N_G(v) \setminus K_4), \quad (3.7)$$

and if every vertex in the common neighbourhood of each leaf has no neighbours outside the tetrahedron and that neighbourhood itself,

$$N_G(w) \subseteq K_4 \cup N_L \quad \forall w \in N_L, \quad (3.8)$$

then u is designated as the internal vertex, v as the external vertex, and s_1, s_2 as the leaf vertices of the tetrahedron.

Under Equations (3.7) and (3.8), the vertices s_1 and s_2 correspond to leaves adjacent to a common internal vertex u , whilst v acts as the connecting vertex between u and the remainder of the graph. The vertices with identical minimal neighbourhood structure correspond to leaves adjacent to a common internal vertex u , whilst the vertex v , having maximal neighbourhood containment, acts as the connecting vertex between u and the remainder of the graph.

For $n = 4$, the vertex u is a binary branching vertex contained in a unique K_4 . In particular, when $n = 5$, the branching vertex u has three leaf children s_1, s_2, s_3 , producing $\binom{3}{2} = 3$ distinct K_4 s

$$\{s_1, s_2, u, v\}, \quad \{s_1, s_3, u, v\}, \quad \{s_2, s_3, u, v\}.$$

More generally, suppose u has k leaf children $s_1, \dots, s_k$, and that these vertices together with v induce a K_{k+2} . Then every pair of leaves (s_i, s_j) determines a distinct $K_4 \{s_i, s_j, u, v\}$ where the total number of such K_4 s generated by this branching configuration is $\binom{k}{2}$. Suppose that s_1, s_2 are chosen without loss of generality, then if the K_4 satisfies Equation (3.7) and Equation (3.8) have

$$x_{s_1u} = x_{us_1} = x_{s_2u} = x_{us_2} = x_{uv} = x_{vu} = 1. \quad (3.9)$$

Proposition 7. *Let s_1, s_2, u, v form a K_4 in square G . If Equation (3.7) and Equation (3.8) are satisfied and*

$$N_G(u) \setminus K_4 = N_G(v) \setminus K_4,$$

then one can choose to assign

$$x_{s_1u} = x_{us_1} = x_{s_2u} = x_{us_2} = x_{uv} = x_{vu} = 1$$

in the tree root H without loss of generality.

Proof. Since s_1, s_2, u, v form a K_4 in G and

$$N_G(u) \setminus K_4 = N_G(v) \setminus K_4,$$

the vertices u and v have identical neighbourhoods outside K_4 . Moreover, since u and v are both adjacent to every vertex of K_4 , it follows that

$$N_G(u) = N_G(v).$$

Thus, exchanging u and v is an automorphism of G . Hence, if a tree root H has v as the internal vertex of the tetrahedral configuration, applying this automorphism allows the roles of u and v to be exchanged without changing the square graph. Therefore, without loss of generality may assume that u is the internal vertex of the tetrahedral configuration and assign

$$x_{s_1u} = x_{us_1} = x_{s_2u} = x_{us_2} = x_{uv} = x_{vu} = 1.$$

□

It remains to verify that whenever a root H of G exists and the conditions in Equation (3.7) and Equation (3.8) are satisfied, imposing the assignment in Equation (3.9) preserves the feasibility of the linear program.

Proposition 8. *Let s_1, s_2, u, v form a 4-clique in square G and Equations (3.7)-(3.8) be satisfied. The linear program with Equation (3.9) never returns an infeasible solution for squares with roots.*

Proof.

□

3.4.3 Single Tree Linear Program

3.5 Tree Proof

Using adjacency matrices, we show that the modified linear program given by still gives integral or fractional solutions for graphs with roots and gives fractional or infeasible solutions for graphs without roots but always return valid integral root solution if there exists a tree root.

3.5.1 Integer Program

Lemma 15. *Consider the graph square G containing vertices i, j and k where edges ik and kj are present and ij is absent in G such that there exists a P_2 . Then every feasible solution of $\mathcal{IP}(G)$ must satisfy*

$$x_{ik} + x_{kj} \leq 1.$$

Proof. As ij is absent in G , $g_{ij} = 0$ such that Equation (2.14) gives $z_{ij} \leq 0$ and Equation (2.15) gives $z_{ij} \geq 0$ such that $z_{ij} = 0$. Therefore Equation (2.13) gives

$$0 = \sum_{a=1}^n y_{iaj} + x_{ij} = \sum_{a=1}^n x_{ia}x_{aj} + x_{ij},$$

where the second equality follows from the product constraints Equations (2.16)–(2.18) together with the integrality of $x \in \{0, 1\}$ by Equation (2.9) and Equation (2.19). Since all variables are non-negative, 0 only holds when $x_{ij} = 0$ and all of $x_{ia}x_{aj} = 0$ and using the integrality of x , this means that at most one of x_{ia}, x_{aj} is equal to 1 and the other must be 0, which can be equivalently written as $x_{ia} + x_{aj} \leq 1$. Taking $a = k$ gives $x_{ik} + x_{kj} \leq 1$ as required. $\square$

3.5.2 Linear Program

Lemma 16. *Consider the graph square G containing vertices i, j and k where edges ik and kj are present and ij is absent in G such that there exists a P_2 . Then every feasible solution of $\mathcal{LP}(G)$ must satisfy*

$$x_{ik} + x_{kj} \leq 1.$$

Proof. Since $g_{ij} = 0$, Equations (??)–(??) imply that $z_{ij} = 0$. Equation (??) therefore gives

$$0 = z_{ij} = \sum_{a=1}^n y_{iaj} + x_{ij}.$$

Since all variables are nonnegative by Equations (3.1)–(3.3), every term in the above sum must be equal to zero with $x_{ij} = 0$ and $y_{iaj} = 0$ for all a . Thus applying Equation (??) to $y_{ikj} = 0$ gives

$$0 = y_{ikj} \geq x_{ik} + x_{kj} - 1,$$

and hence $x_{ik} + x_{kj} \leq 1$ as required. $\square$

3.5.3 Max Linear Program

Conjecture 1. *Let x be an optimal solution to $\mathcal{MLP}(G)$. If edge $ij \notin E_G$, that is $g_{ij} = 0$, then for every $k \in \{1, \dots, n\}$,*

$$x_{ik} + x_{kj} \in \{0, 1\}.$$

Proof. $\square$

Chapter 4

Further Work

In this chapter we discuss additional work, further concepts and interesting ideas around specific structural properties that square intrinsically holds. Most of these have not been rigorously proven, with only sketch proofs given or intuitions given behind why they help improve results.

Further work would naturally be to improve many of these nascent ideas and develop formal proofs.

4.1 Symbolic Groups

Lemma 15 and its linear relaxation equivalent Lemma 16 give extremely strong structure into how the linear program operates. Using the maximization objective function Lemma 16, any two edges ij, jk which exist in G but ki , can be assigned the weights a, b respectively where $x_{ij} = a, x_{jk} = b$ and $a + b \leq 1$.

Under certain conditions, the relationship can hold with equality $a + b = 1$ which therefore gives a methodology for tackling scenarios where the maximal fractional objective function is greater than the optimal integer objective function corresponding to a valid square root. By continuously iterating Lemma 16 can eventually categorize and formulate equations for many of the weights on the various edges.

Consider this graph square on 6 vertices with three different roots

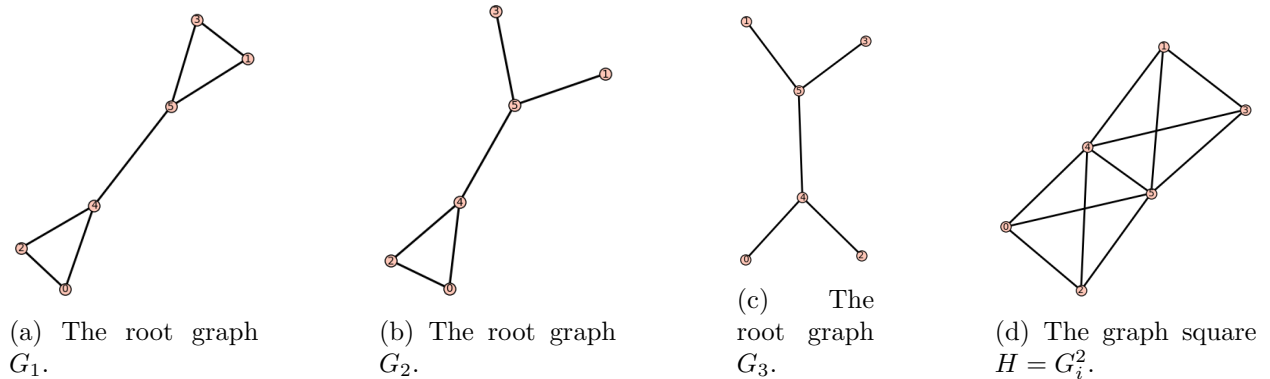


Figure 4.1: Three non-isomorphic square roots of the same six-vertex graph square.

G_1 is the unique maximal root. Assigning $a = x_{14}$ gives using Lemma (??) that $1 - a = x_{40}$ since 01 does not exist in G . The same applies such that $1 - a = x_{42}$ and $a = x_{43}$, whereby the

existence of edges 13 and 02 prevent additional constraints being imposed upon a , it is internally consistent. By symmetry can likewise assign $b = x_{15} = x_{35}$ and $1 - b = x_{50} = x_{52}$. Setting $a = 0$ and $b = 1$ thus gives x_{40}

Consider this graph square on 6 vertices with only one unique root

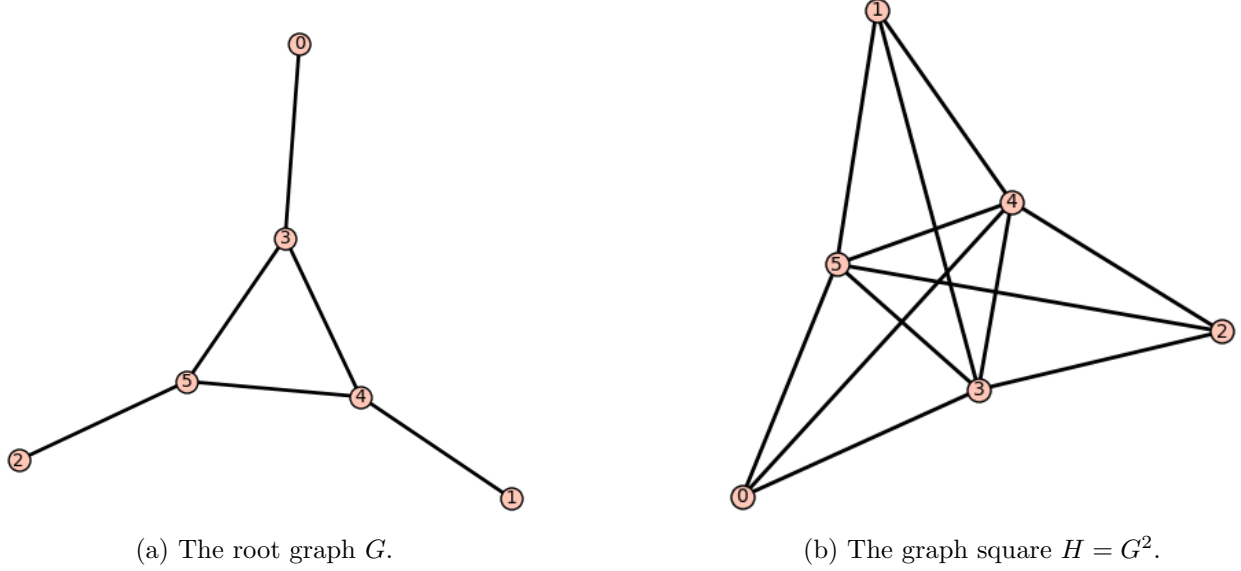


Figure 4.2: A six-vertex graph and its square.

Assigning $a = x_{03}$ gives using Lemma (??) that $x_{31} = 1 - a$ since 01 does not exist in G , and likewise with 12 does not exist in G have $x_{32} = a$. However with the absence of 02, must have $x_{03} + x_{32} \leq 1$ such that $a \leq \frac{1}{2}$. By symmetry this likewise holds for $b = x_{05} = x_{15}$, $1 - b = x_{52}$ and $c = x_{42} = x_{41}$, $1 - c = x_{40}$ giving $b, c \leq \frac{1}{2}$.

In this case, rather than maximizing the objective function and giving a fractional solution, setting $a = b = c = 0$ gives a valid integral solution.

4.2 Vertex Deletion

As single triangular decomposition gives perfect characterizations of the root using the linear program for up to $n = 5$, a further refinement is obtained by analyzing fractional solutions through vertex deletion and by considering the subgraphs after this process.

Let G^- be the subgraph of G defined by $(V^- = V - \{v\}, E_{G^-} = E_G \setminus \{uv \mid u \in V^-\})$ and if it has a valid square root denote it by H^- . Let H_- be the subgraph of the root graph H satisfying the relationship $H^2 = G$ defined by $(V^-, E_H \setminus \{uv \mid u \in V^-\})$. The idea is that if G^- has a valid square root, can construct H_- from H^- by considering the following scenarios for vertex v

- i) v is connected to the subgraph V^- by only one edge to $u \in V^-$
- ii) v is connected to the subgraph V^- by two edges to $u, w \in V^-$ where $uw \in E_{H_-}$
- iii) v is connected to the subgraph G' by two edges to vertices $u, w \in G'$ where $uw \notin E_H$ and there is no P_2 between u, w in G' .

- iv) v is connected to the subgraph G' by two edges to vertices $u, w \in G'$ where $uw \notin E_H$ and there is a P_2 between u, w in G' .

where the remaining possibilities have the above sub-cases.

- i) If G^- has a valid square root H' , then $H = H' + \{v\}$ where $E_H = E_{H'} \cup vu$.
- ii) If G' has a valid square root H' , then $H = H' + \{v\}$ where $E_H = E_{H'} \cup vu$ or $E_H = E_{H'} \cup vw$ ($E_H = E_{H'} \cup vu, vw$ also satisfies but is not necessary)
- iii)

If the solution of the original linear program is fractional, starting from the vertex $v \in V$ with the smallest degree, the linear program is ran on the induced subgraph $G' = V - \{v\}$. If the resulting solution is integral, then the fractional behaviour of G is localized at v and the algorithm attempts to lift the integral structure of G' back to G by enforcing that v is attached to one of its neighbours. For each $u \in N_G(v)$, the constraint

$$x_{vu} = 1, \quad x_{uv} = 1, \tag{4.1}$$

is added. If the resulting solution is integral, then v is identified as a leaf vertex adjacent to u in the square root, and the solution is accepted.

If no such u yields an integral solution, the process is repeated over all vertices. This technique runs in polynomial time, as it performs at most n vertex-deletion checks and, for each such vertex v , at most $\deg_G(v) \leq n - 1$ constrained linear program solves. Since each linear program has $O(n^2)$ variables and constraints and can be solved in polynomial time $T_{LP}(n)$, the overall procedure requires at most $O(n^2 \cdot T_{LP}(n))$ time, and hence remains polynomial in n .

Proposition 9. *Vertex deletion completely categorizes all proper interval roots in polynomial time.*

Proof. □

4.3 Duality

Consider the dual formulation of the max objective linear program defined in Section 3.3 by first restating it in canonical form.

Objective Function:

$$\min - \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \tag{4.2}$$

Variables:

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (4.3)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (4.4)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (4.5)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} = 0, \forall i \in \{1, \dots, n\}, \quad (4.6)$$

$$z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} = 0, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (4.7)$$

$$(2) \text{ Indicator function: } z_{ij} - M g_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (4.8)$$

$$-z_{ij} + g_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (4.9)$$

$$(3) \text{ Product: } y_{ikj} - x_{ik} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (4.10)$$

$$y_{ikj} - x_{kj} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (4.11)$$

$$x_{ik} + x_{kj} - y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (4.12)$$

$$(4) \text{ Symmetry: } x_{ij} - x_{ji} = 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (4.13)$$

$$(5) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}, \quad (4.14)$$

$$(6) \text{ Variable Bound: } x_{ij} - 1 \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (4.15)$$

$$y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}. \quad (4.16)$$

Its dual can be rewritten as

Objective Function:

$$\max \sum_{i=1}^n \sum_{j=1}^n \left(-M g_{ij} \gamma_{ij} + g_{ij} \delta_{ij} - \lambda_{ij} \right) - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n (\eta_{ikj} + \mu_{ikj}). \quad (4.17)$$

Dual Variables:

$$\alpha_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (4.6)}), \quad (4.18)$$

$$\beta_{ij} \in \mathbb{R}, \quad \forall i \neq j \in \{1, \dots, n\}, (\text{associated with (4.7)}), \quad (4.19)$$

$$\gamma_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (4.8)}), \quad (4.20)$$

$$\delta_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (4.9)}), \quad (4.21)$$

$$\epsilon_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (4.10)}), \quad (4.22)$$

$$\zeta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (4.11)}), \quad (4.23)$$

$$\eta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (4.12)}), \quad (4.24)$$

$$\theta_{ij} \in \mathbb{R}, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (4.13)}), \quad (4.25)$$

$$\iota_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (4.14)}), \quad (4.26)$$

$$\lambda_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (4.15)}), \quad (4.27)$$

$$\mu_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (4.16)}). \quad (4.28)$$

Dual Constraints:

$\forall i, j \in \{1, \dots, n\}$, each primal x_{ij} :

$$\begin{cases} -1 - \sum_{k=1}^n \epsilon_{iik} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iik} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \geq 0, & \text{if } j = i, \\ -1 + \alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \theta_{ij} - \theta_{ji} + \lambda_{ij} \geq 0, & \text{if } j \neq i, \end{cases} \quad (4.29)$$

$\forall i, k, j \in \{1, \dots, n\}$, each primal y_{ikj} :

$$\begin{cases} -\alpha_i + \epsilon_{iki} + \zeta_{iki} - \eta_{iki} + \mu_{iki} \geq 0, & \text{if } j = i, \\ -\beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \geq 0, & \text{if } j \neq i, \end{cases} \quad (4.30)$$

$\forall i, j \in \{1, \dots, n\}$, each primal z_{ij} :

$$\begin{cases} \alpha_i + \gamma_{ii} - \delta_{ii} = 0, & \text{if } j = i, \\ \beta_{ij} + \gamma_{ij} - \delta_{ij} = 0, & \text{if } j \neq i. \end{cases} \quad (4.31)$$

whereby the Lagrangian is

$$\begin{aligned} \mathcal{L}(x, y, z) = & - \sum_{i=1}^n \sum_{j=1}^n x_{ij} + \sum_{i=1}^n \alpha_i \left(z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} \right) + \sum_{i \neq j} \beta_{ij} \left(z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} \right) \\ & + \sum_{i=1}^n \sum_{j=1}^n \gamma_{ij} (z_{ij} - M g_{ij}) + \sum_{i=1}^n \sum_{j=1}^n \delta_{ij} (-z_{ij} + g_{ij}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \epsilon_{ikj} (y_{ikj} - x_{ik}) \\ & + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \zeta_{ikj} (y_{ikj} - x_{kj}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} (x_{ik} + x_{kj} - y_{ikj} - 1) \\ & + \sum_{i=1}^n \sum_{j=1}^n \theta_{ij} (x_{ij} - x_{ji}) + \sum_{i=1}^n \iota_i x_{ii} + \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} (x_{ij} - 1) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj} (y_{ikj} - 1). \end{aligned}$$

Collecting terms gives

$$\begin{aligned} \mathcal{L}(x, y, z) = & \sum_{i=1}^n \sum_{j=1}^n x_{ij} \left[-1 + \mathbf{1}_{\{j=i\}} \left(\alpha_i - \sum_{k=1}^n \epsilon_{iki} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iik} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \right) \right. \\ & \left. + \mathbf{1}_{\{j \neq i\}} \left(\alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \lambda_{ij} \right) + \theta_{ij} - \theta_{ji} \right] \\ & + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n y_{ikj} \left[-\mathbf{1}_{\{j=i\}} \alpha_i - \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \right] \\ & + \sum_{i=1}^n \sum_{j=1}^n z_{ij} \left[\mathbf{1}_{\{j=i\}} \alpha_i + \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \gamma_{ij} - \delta_{ij} \right] \\ & - \sum_{i=1}^n \sum_{j=1}^n M g_{ij} \gamma_{ij} + \sum_{i=1}^n \sum_{j=1}^n g_{ij} \delta_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} - \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj}. \end{aligned}$$

4.4 Rounding

Rounding fractional solutions such that an integral solution or the problem is deemed infeasible gives the classification in Figure 9 is obtained.

Result Diagrams

Integer Program

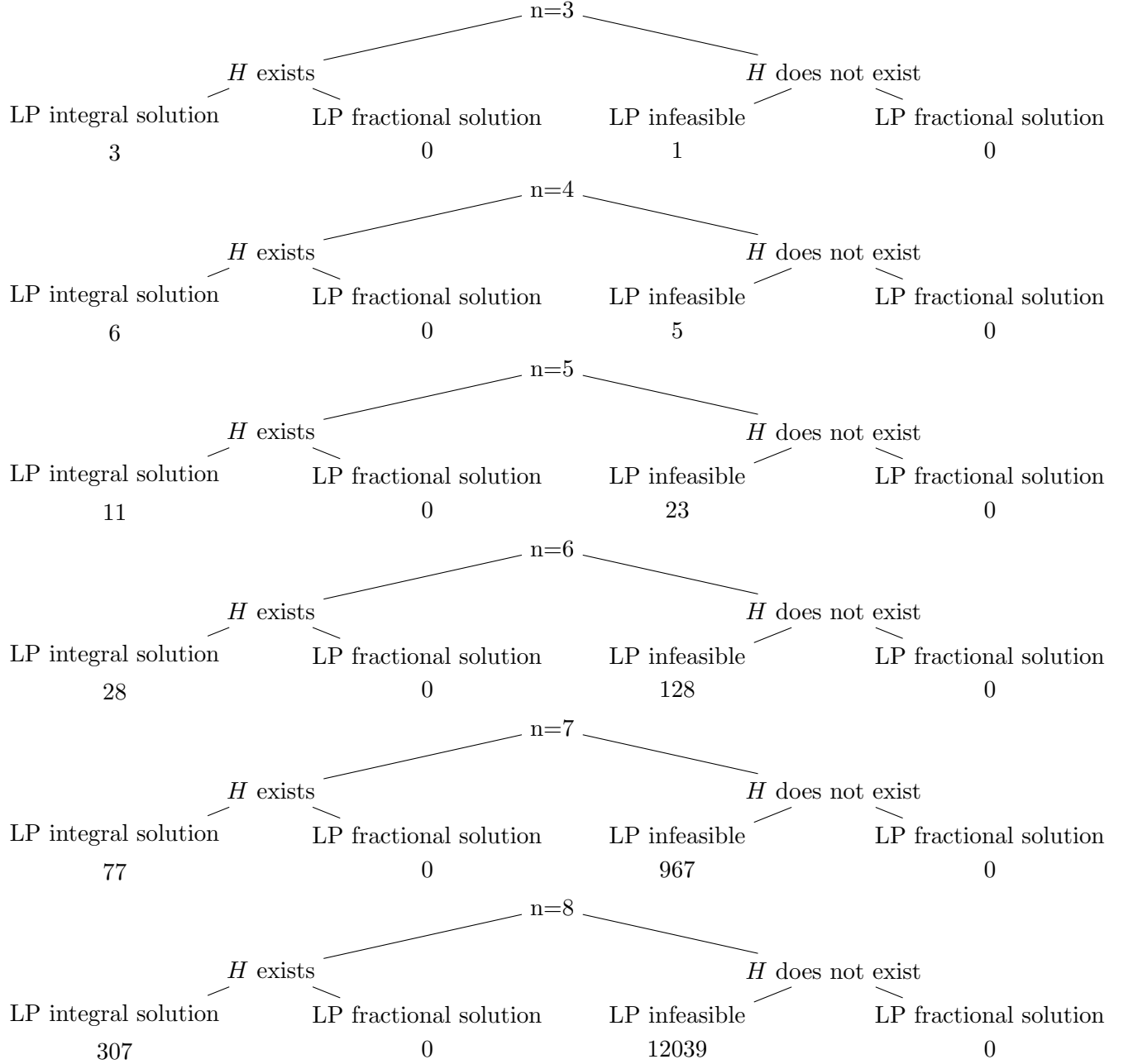


Figure 3: IP outcomes for different values of n on integer program.

1

¹The total number of simple graphs on $n = 2, 3, 4, \dots$ unlabeled nodes is the sequence 2, 4, 11, 34, 156, 1044, 12346, ... given by A000088.

The IP integral solution sequence is the total number of graphs which are squares is the sequence 2, 3, 6, 11, 28, 77, 307, ... given by A382181.

Linear Program

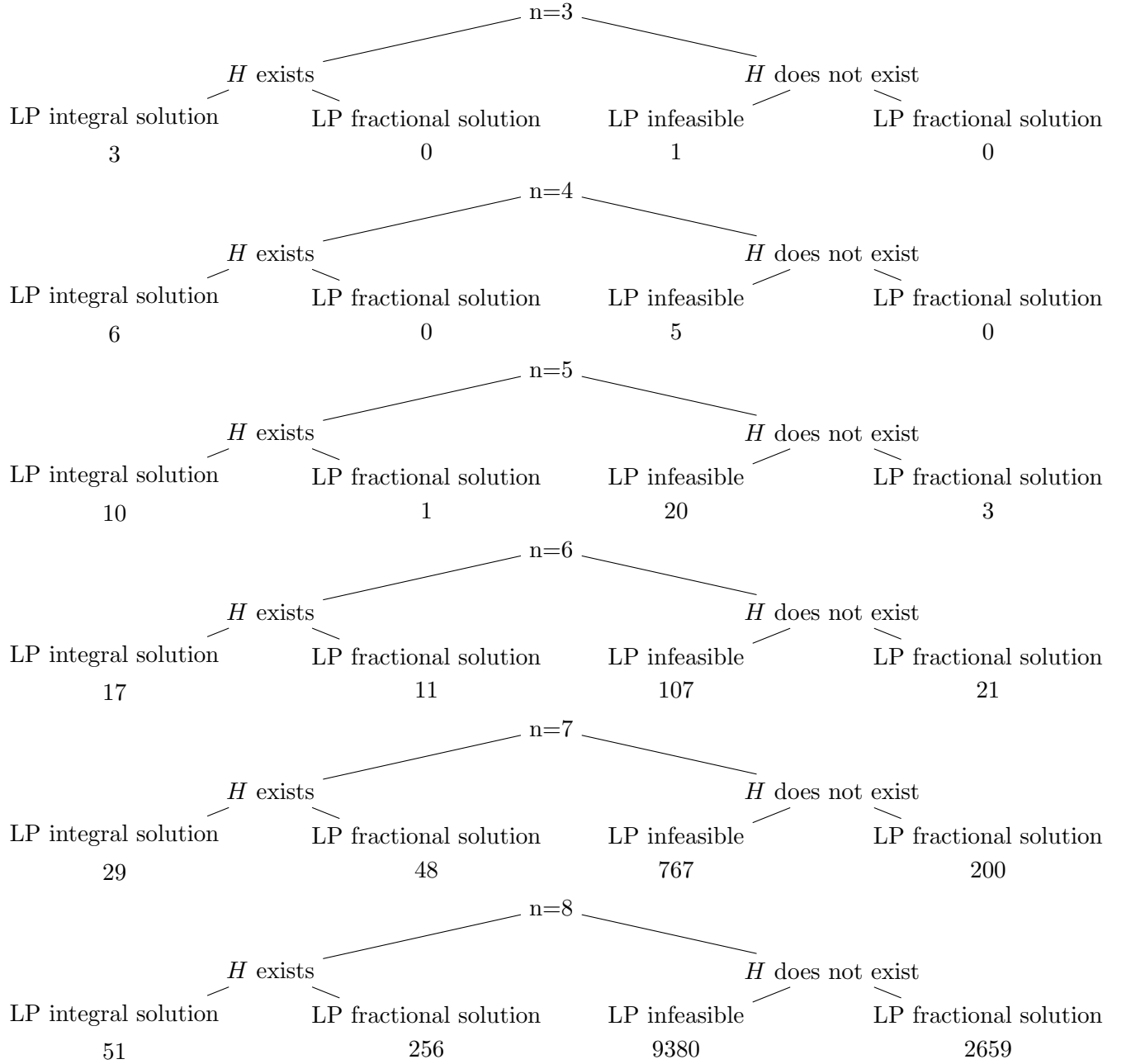


Figure 4: LP outcomes for different values of n on base linear program.

2

²The LP integral solution sequence 2, 3, 6, 10, 17, 29, 51, ... is given by A285665.

Max Objective Function

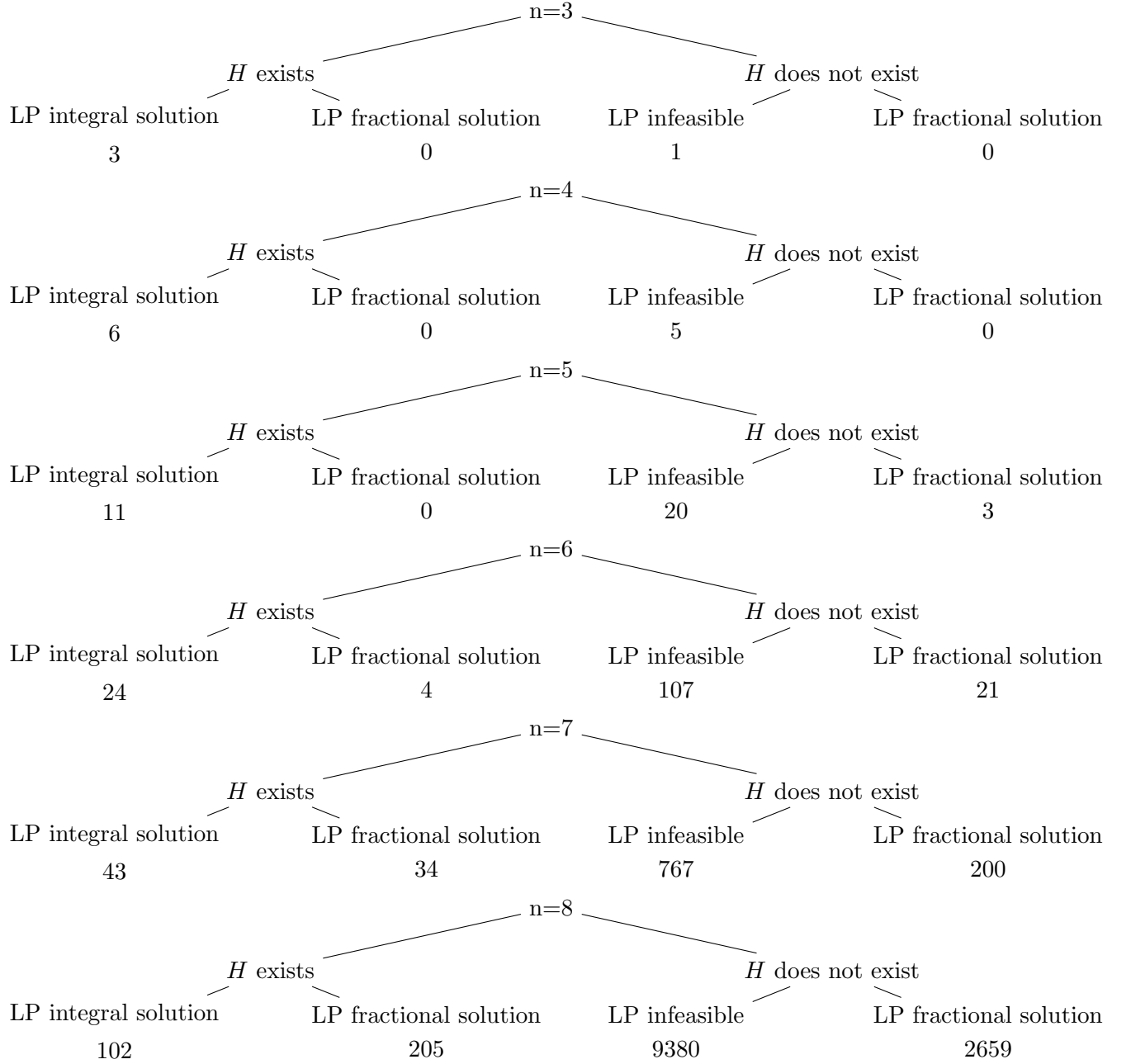


Figure 5: LP outcomes for different values of n on the linear program with max objective function.

3

³The LP integer solution sequence 3, 6, 11, 22, 42, 102, ... is very similar to the number of distinct characteristic polynomials of trees with n nodes - A226594.

Min Objective Function

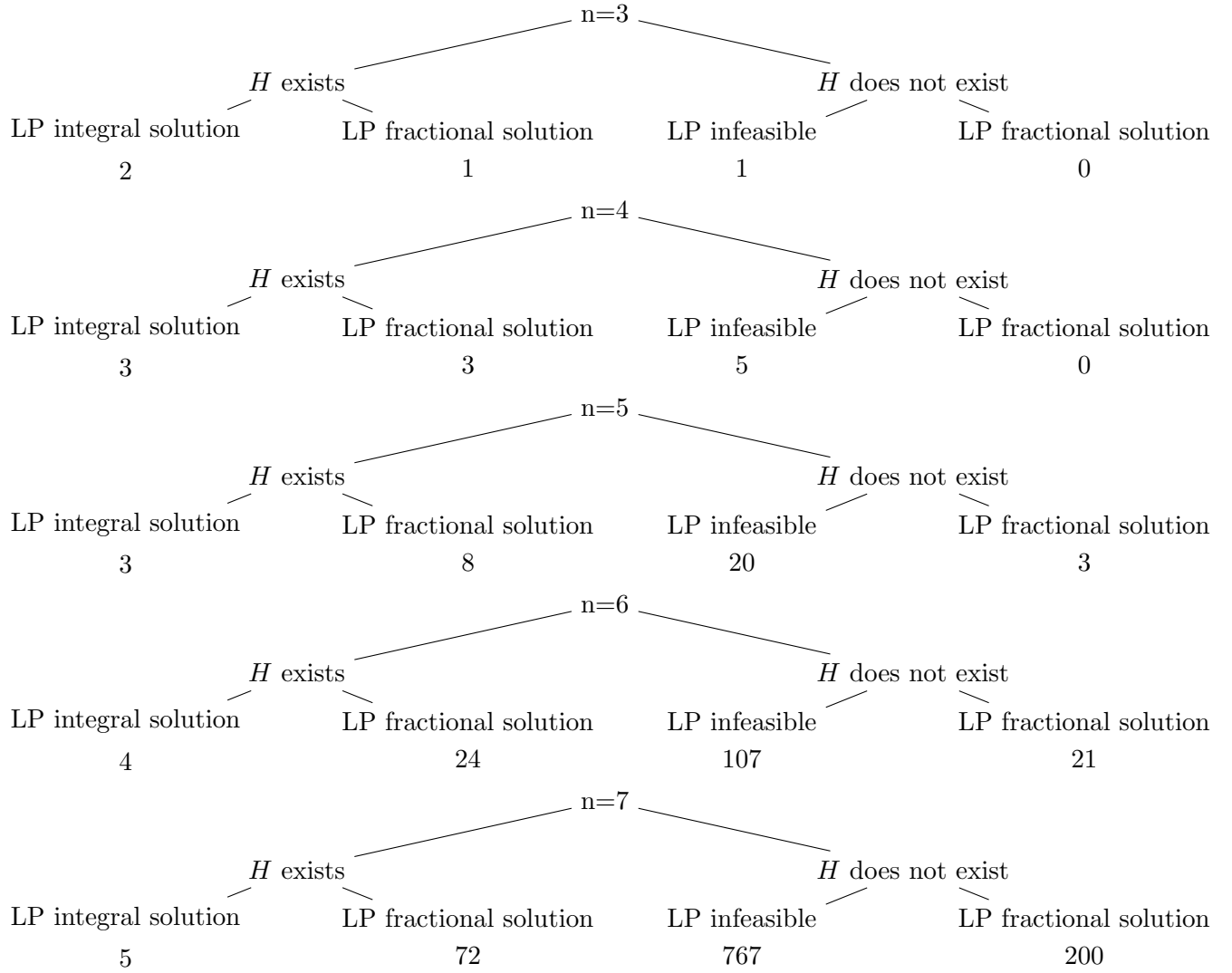


Figure 6: LP outcomes for different values of n on the linear program with min objective function.

Single Triangular Decomposition

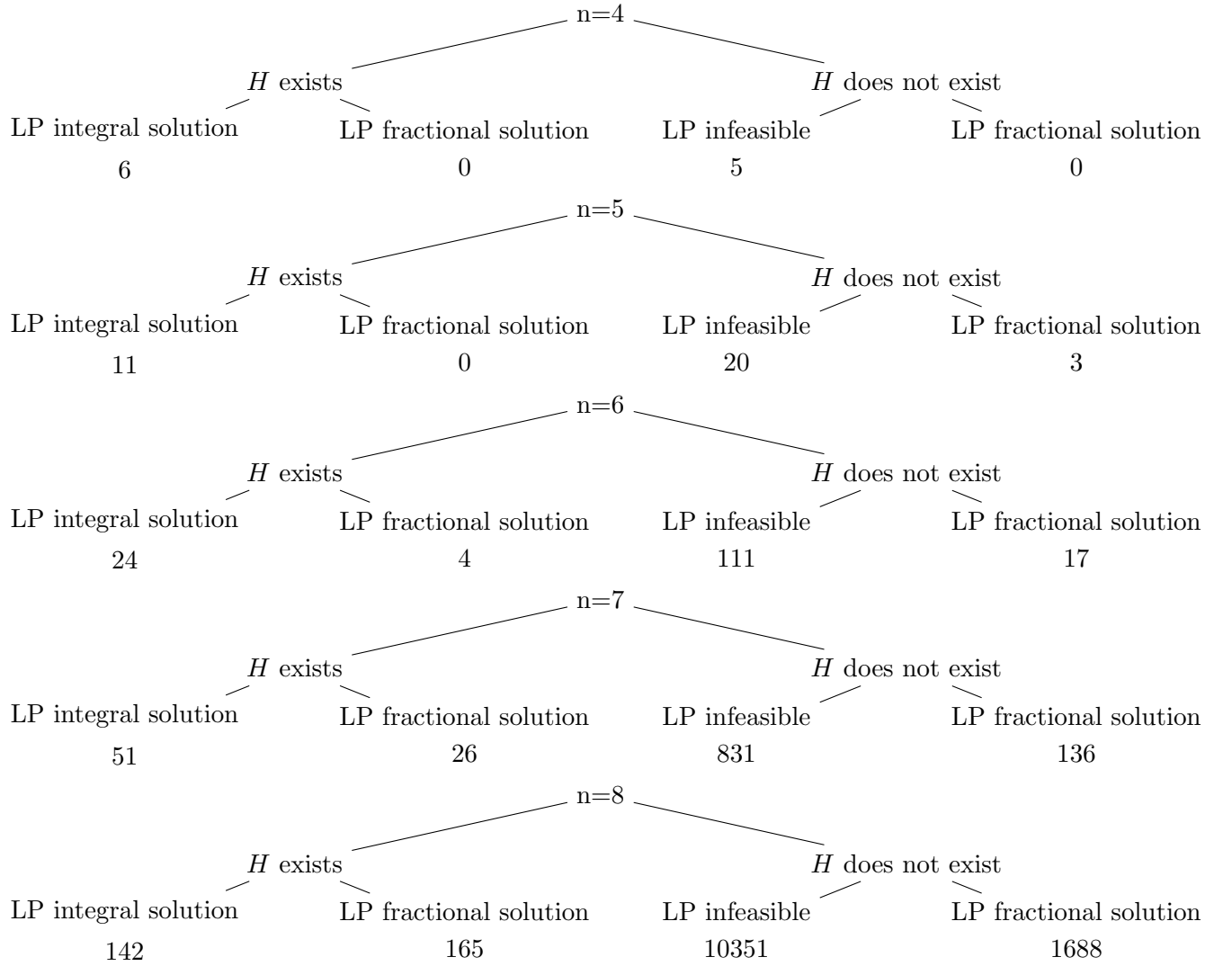


Figure 7: LP outcomes for different values of n on the linear program with max objective function and single triangular decomposition.

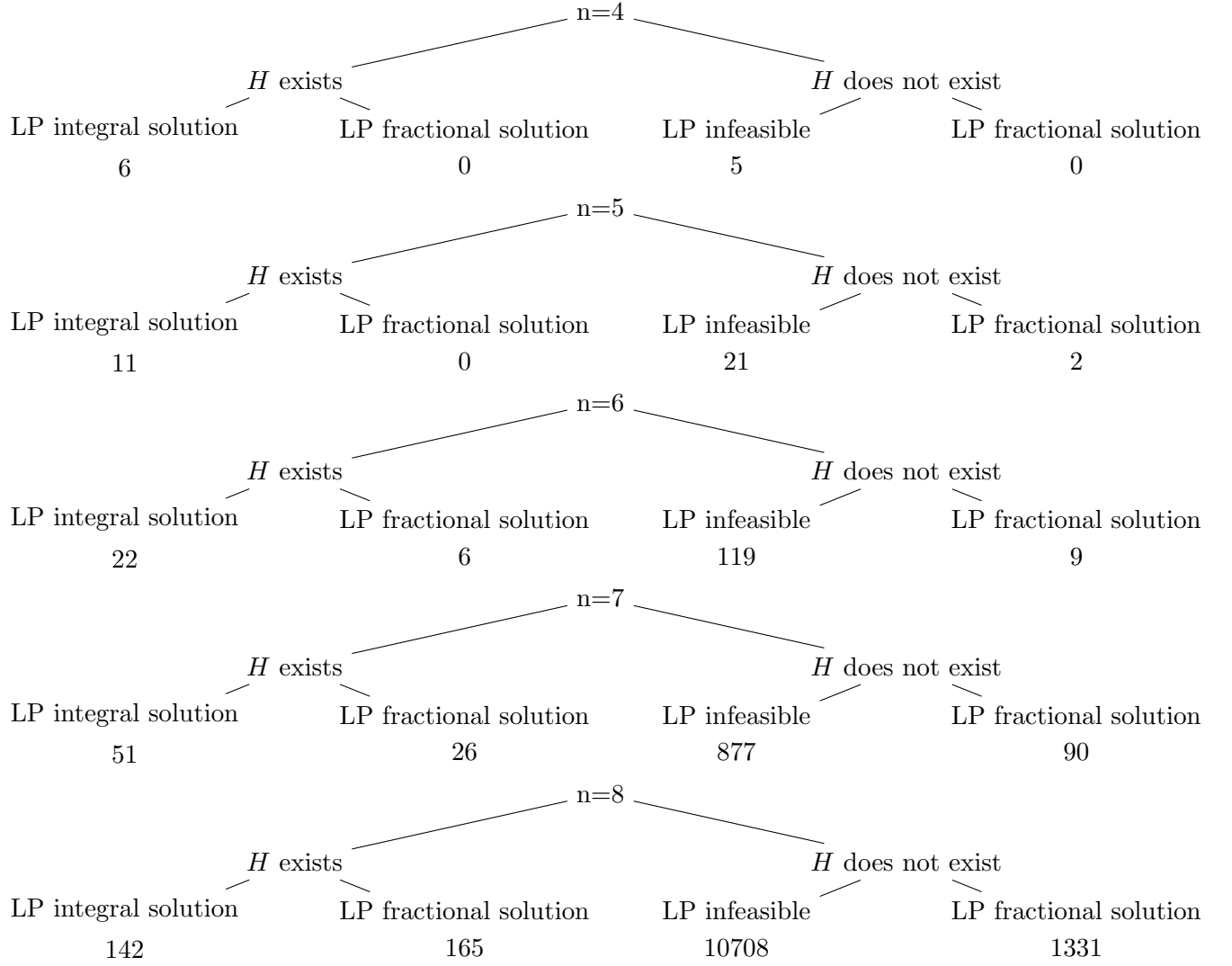


Figure 8: LP outcomes for different values of n on the linear program with max objective function and single triangular decomposition with recursion for tie-breaking.

Rounding

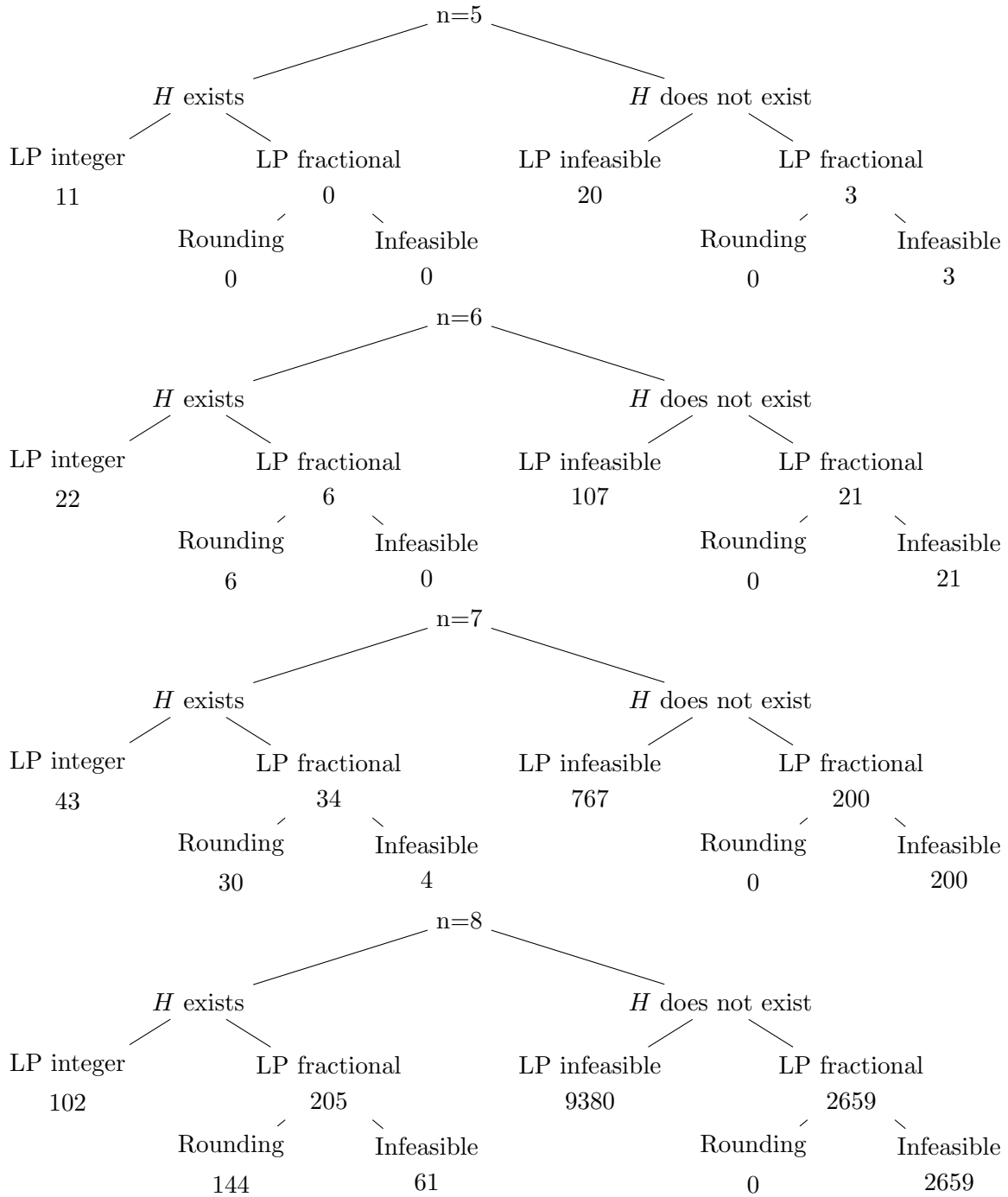


Figure 9: LP outcomes for different values of n on the linear program with max objective function and using rounding for fractional solutions.

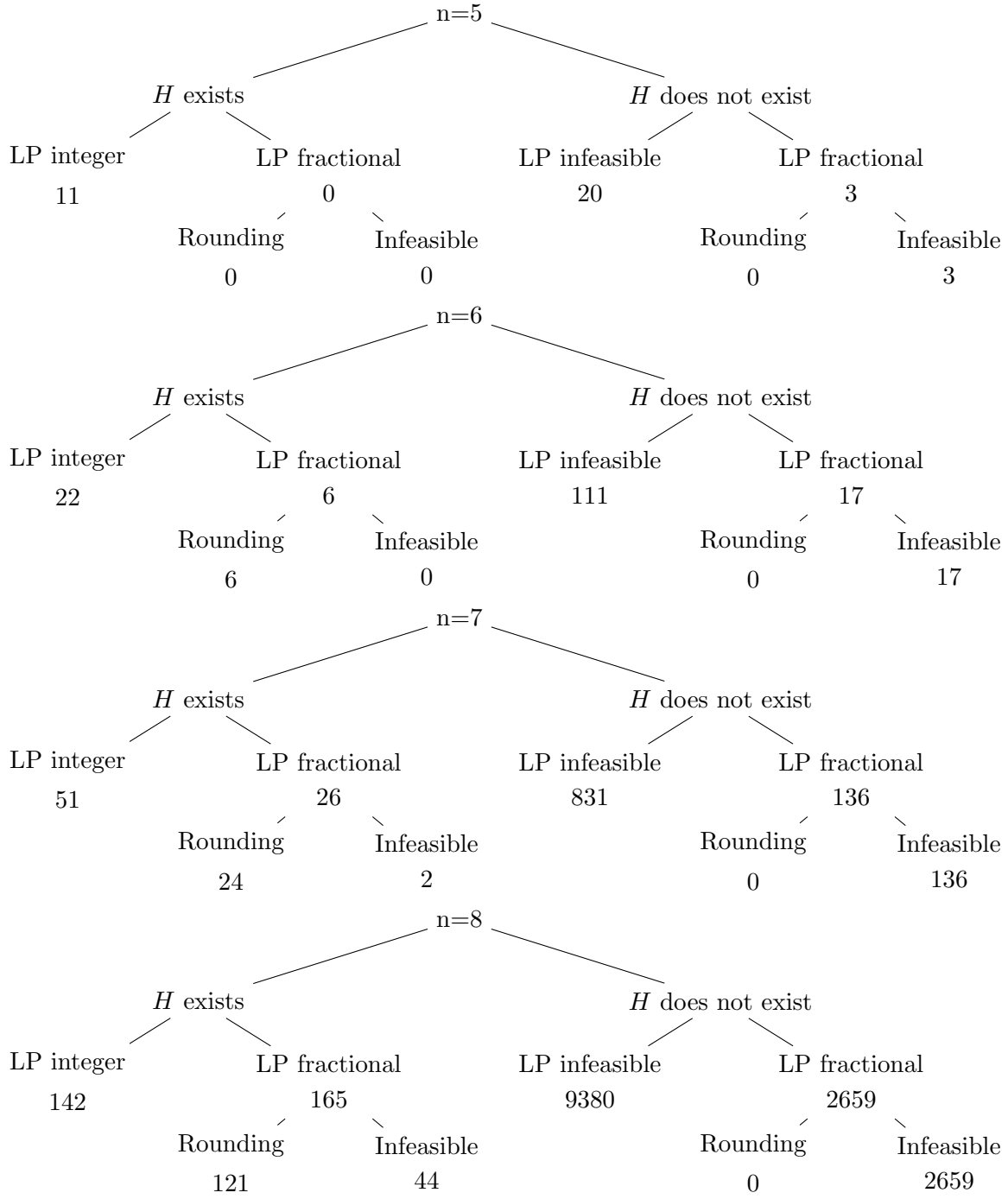


Figure 10: LP outcomes for different values of n on the linear program with max objective function, single triangular decomposition and using rounding for fractional solutions.

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